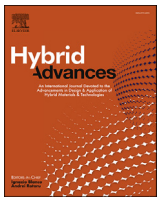
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Advances in machine learning-aided design of reinforced polymer composite and hybrid material systems



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ABSTRACT

Reinforced composite is a preferred choice of material for the design of industrial lightweight structures. As of late, composite materials analysis and development utilizing machine learning algorithms have been getting expanding consideration and have accomplished extraordinary upgrades in both time productivity and expectation exactness. This review encapsulates recent advances in machine learning-based design of reinforced composite during the last half-decade. It summarizes the limitations of traditional methods of reinforced composite development and presents a detailed protocol of machine learning in composite materials technology; implementation of machine learning algorithms in reinforced composite material design was covered, with an emphasis on the importance of data hygiene. Machine learning integration in material and process selection, and data sourcing techniques for machine learning-based design were also examined. The evaluation also looked at emerging digital tools and platforms for implementing machine learning algorithms. In addition, an essential effort was made to identify research gaps and define areas for further research. This review is indeed designed to provide some direction for future research into the use of machine learning for composite material design.

1. Introduction

Composites are made up of at least two materials having superior quality to that of the independent units. The reinforcing phase is composed of nanoparticles or fibers, whereas the matrix phase usually consists of formable resin. The area is defined as the point where the reinforcement and matrix phases intersect. The mechanical interaction of the reinforcement and the resin govern the composite qualities, whilst chemical, electrical, and thermal characteristics are controlled by resin frameworks. Resilience, spatial solidity, and heat resistance are on the other hand provided by the strands/reinforcement. Plasticizers affect a variety of qualities, including tone and outer layer smoothness, as well as durability and fire resistance. The handling process, additives, the sort, sum, and organization of the resin frameworks and fortifications dictate the final properties of emerging composite [1]. Within such complexity in polymer composite materials and hybrid systems, an effective design of reinforced composite requires the application of an algorithm.

Hybrid material systems are designed by the specific combination of more than two different materials/different forms of reinforcement at the nanometer or molecular level [2]. In contrast to individual reinforced polymer composites, a hybrid material system—also known as a composite made from two or more types of fibers—produces materials that are stiffer and stronger. In most hybrid material systems, one type of fiber, has a low modulus and/or cheap cost, whereas the other type, has a higher modulus and/or greater cost [3]. When multiscale components are combined, the resultant hybrid material systems typically gain new properties, and these qualities may be tuned by the unique chemical and physical properties of individual parts, structure, and interfaces between

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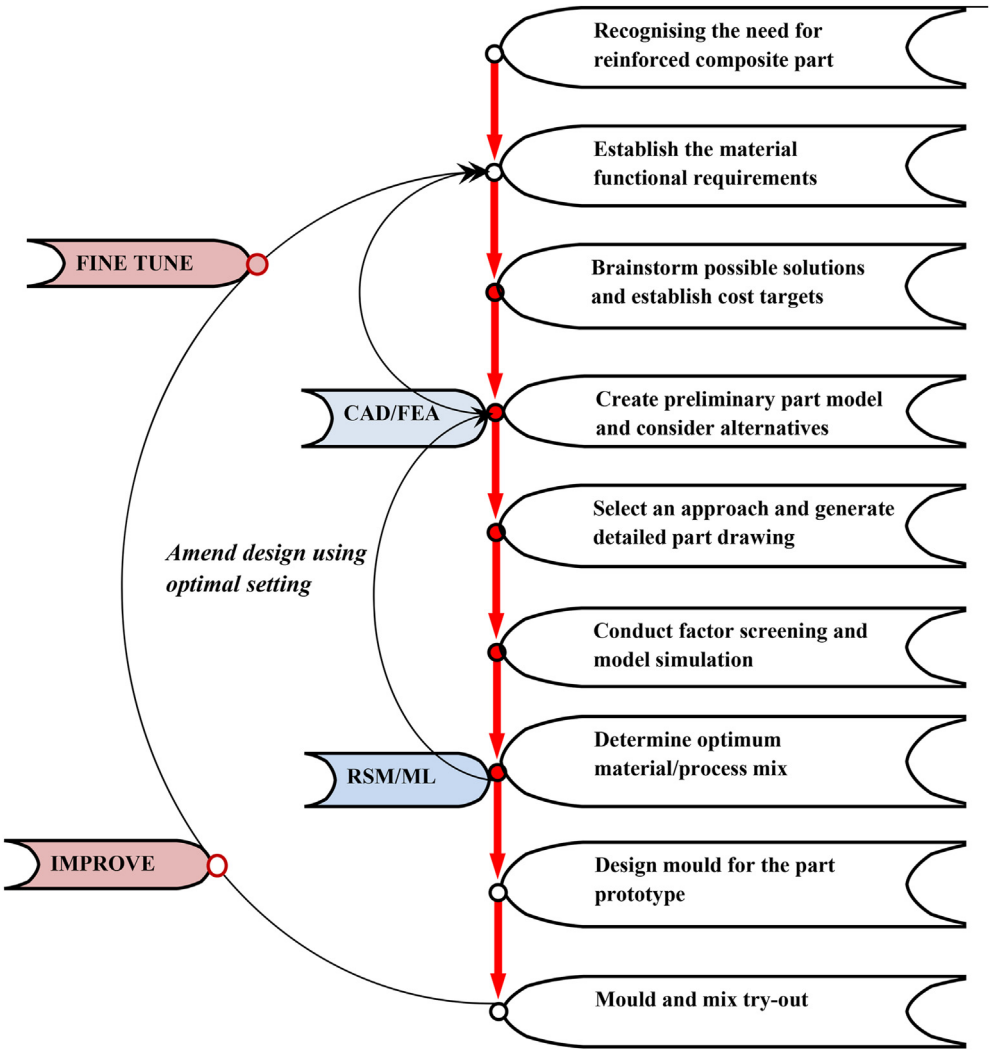


Fig. 1. Flow of reinforced composite material part design process showing the stage for Response surface methodology (RSM), Machine Learning (ML), Computer Aided Design (CAD) and Finite element analysis (FEA) deployment.

individual systems. For lightweight, high strength and high rigidity, complex shaped structural components, which are used in the automotive industry and in aerospace, hybrid material systems offer an outstanding potential [4–6]. A comprehensive understanding regarding the interaction of the individual components of the hybrid material is of great importance for a more efficient design of future structures.

Reinforced polymer composite and hybrid material systems offers a significant design flexibility; hence such materials are often used in aviation, construction, transportation, power, building, nautical, defense, and sport and leisure. The unending application of these innovative composite materials relies upon the creative mind of the design engineers as they grow much more imaginative applications. Machine Learning (ML), as an approximation tool offers a potential way to use available datasets and infer data-driven models, which may then be used to speed up the identification of innovative composite materials [7]. ML has transformed numerous elements of reinforced composite development during the last half-decade. ML modeling approaches have been utilized to speed up the design of reinforced composite structures, which has been a remarkable success [8–10].

ML is a branch of Artificial Intelligence that utilizes a scope of factual and probabilistic methodologies, permitting computers to learn and recognize hidden designs (input-output connections) from enormous and regularly noisy data collections. ML is presently seen as an effective methodology for the design and finding of novel composite materials for a wide assortment of utilizations [11–14]. ML, therefore, is the process of enabling computers to handle tasks that are hitherto been completed by individuals. It centers around the advancement of computer programs that can receive data and use it to learn on their own [15].

The goal of ML is to create algorithms that can recognize arrangements in complicated datasets automatically via a learning technique and then utilize these arrangements to make predictions. Because of its high prediction performance and relatively low processing cost, ML has been widely employed in the prediction of properties, discovery of novel materials, microstructure-based material design, stress analysis [16], topology optimization of reinforced composites [17] and optimization of fiber-reinforced polymer composite components [18]. Despite the widespread use of ML and its influence in recent years, there has been little investigations into how it may be used to design robust composite materials.

The purpose of this review is to appraise recent developments in machine learning-based design of reinforced composites and hybrid systems; we give exceptional consideration to the background of reinforced composite material part design process, the place of data-driven material design, drawbacks of traditional methods of reinforced composite development, ML protocol in the technology of composite materials, implementation of ML algorithms in reinforced composite material design, a general classification of ML algorithms, data hygiene, data cleaning, integration of ML in reinforced composite materials design/

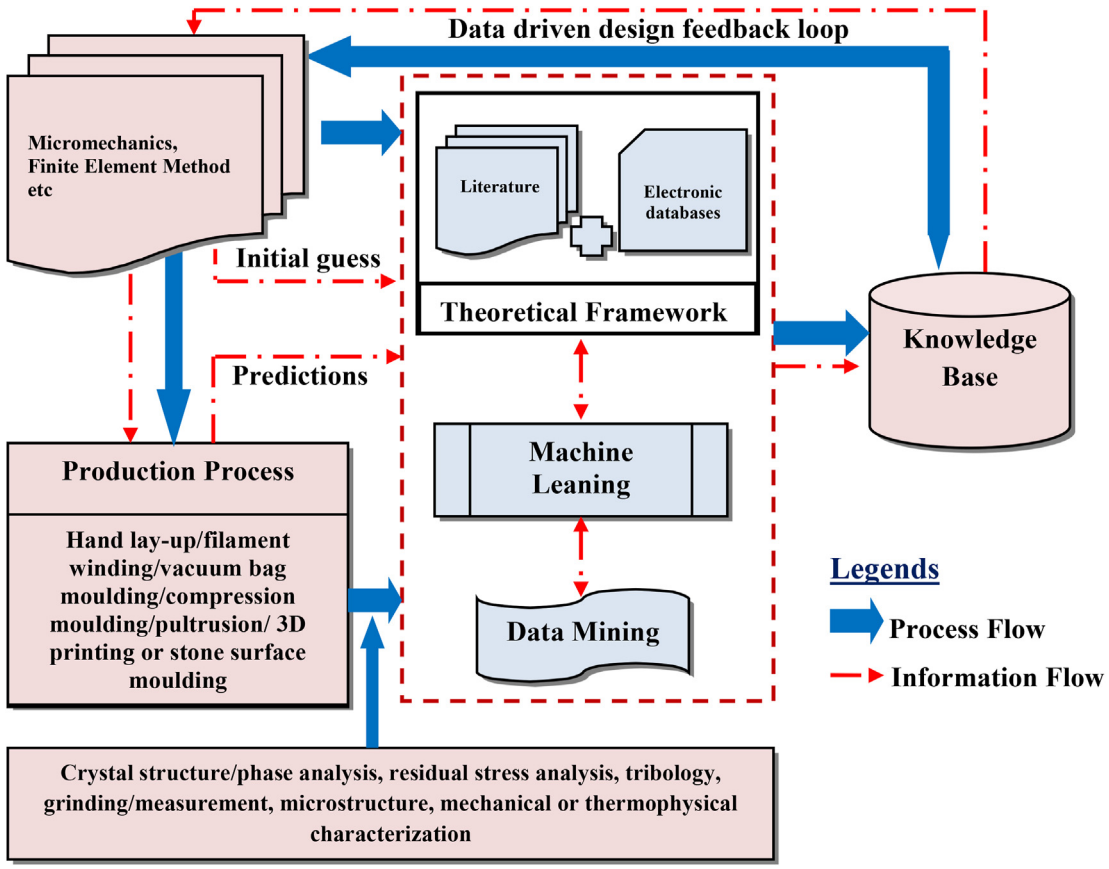


Fig. 2. Data-driven design of composite material showing production routes, data acquisition sources and simulation tools.

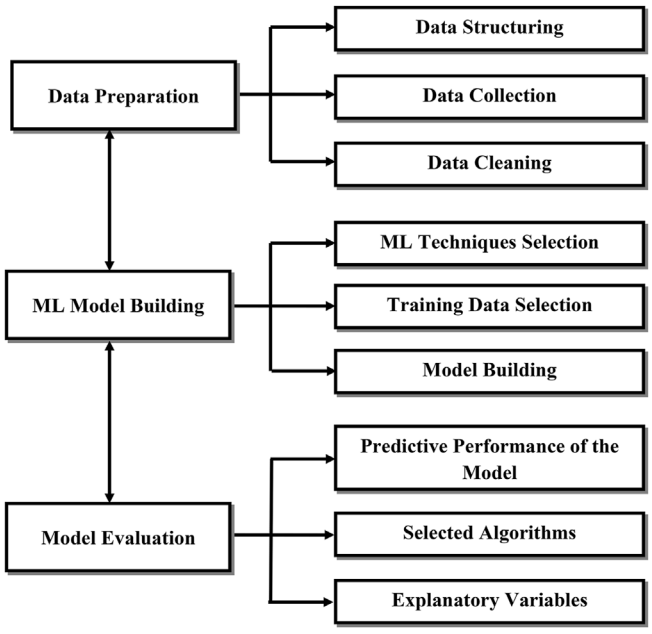


Fig. 3. Steps in the ML process for predicting reinforced composite material properties.

process selection, data sourcing for machine learning-based design, property prediction in reinforced composite development using ML algorithms and detailed appraisal of emerging digital tools and platforms used for implementation of ML algorithms. The present study is beneficial to both experienced and upcoming researchers as it provides a better grasp of the current research and discussions on ML in the field of reinforced composite design.

2. Scope and methodology

The current review encompasses all publications from the recent half-decade that deals with machine learning-based design and analysis of reinforced composites. This was accomplished regardless of the ML technique used or the type of reinforced composite preferred. The articles were found using the Google Scholar search engine, which allows users to access both physical and electronic copies of academic material in several publishing formats and subjects. Google Scholar search engine was chosen because it searches broadly and numerous databases allow Google Scholar to see their materials.

The index incorporates most peer-reviewed journals, books, conference proceedings, preprints, abstracts, specialized reports, theories, expositions, and other insightful publications on machine learning-aided design of reinforced composite material. The search was guided by keywords, and quotes were used to find a precise match. The advanced search interface choices in the left-hand panel were utilized to narrow the search results even more by limiting the years covered by the search to 2018–2022 and excluding patents. The deep web, which is contained on the World Wide Web that is not indexed by ordinary search engines was not covered.

3. Background to reinforced composite material part design process

Due to the adaptability of reinforced composite and the numerous material and process decisions involved, scholars suggest that new composite material developers should work together with mold makers

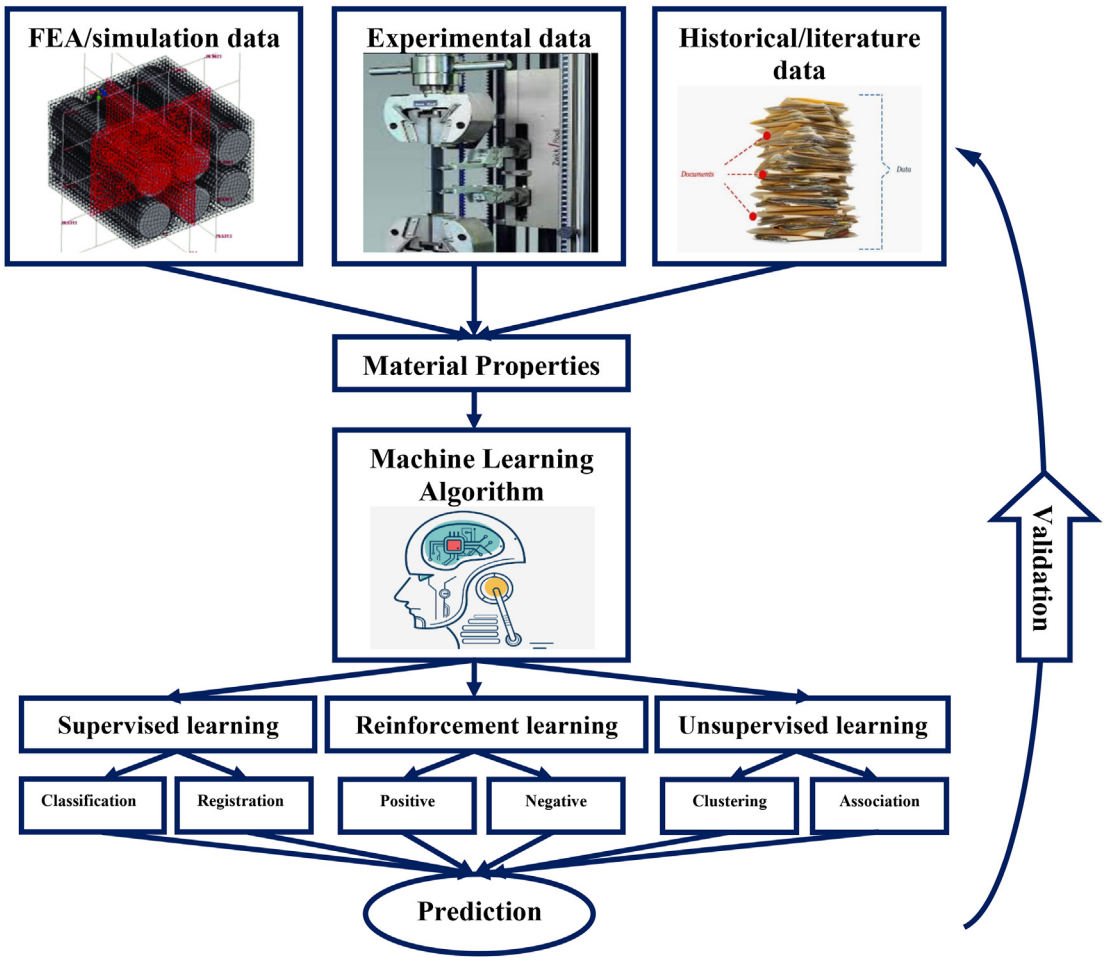


Fig. 4. The protocol of ML **in reinforced composite design**.

and material providers for proper selection of reinforcement (natural or synthetic fiber), resin framework (thermoset or thermoplastic) and preparing strategy (hand-layup, etc) [19–21]. This is because the mechanical strength of every reinforced composite relies upon the volume fraction and arrangement of the reinforcement within the matrix; while thermal, electrical, and chemical properties rely upon the constitution of the matrix framework; and the molding process is controlled by size, shape, parts intricacies and conditions on manufacture [22,23]. Overall, the effectiveness of the favorable design is dependent on a prudent choice of materials/process predicated on a clear recognition of need as shown in Fig. 1.

The composite design process is generally initiated with the identification of the need for the reinforced composite part or with the recognition of a potential market for the composite product and application [24–26]. Acknowledgment of the need necessitates a great deal of creativity and inventiveness because the need is for the most part not usually apparent and can be obscure. The need brings forth different strides in the reinforced composite design process, it follows that if there is no need, there will not be any motivation to begin the nitty-gritty, tedious, and exceptionally complex issue surrounding reinforced composite material design activities.

The second stage involves the establishment of functional requirements which incorporate surface quality and appearance, stability, and compatibility with alternative materials. It covers the mechanical, physical, and chemical properties of the material. Other function-related prerequisites may include match inability and tribological characteristics. The establishment of functional requirements is followed by brainstorming. Designers often explore several alternatives before commencing a design which precedes accumulation a list of possible alternatives [27]. Also cost estimation and optimization are done as soon as feasible and as precisely as possible. Many authors [28,29] have reported the benefits of early-stage cost analysis in the reinforced composite design process. Such cost considerations cover the recurring cost (which includes energy, direct labor, and material costs) and non-recurrent costs (which include facility, tooling, indirect labor, and equipment). Finishing, packaging, storage, quality control, inventory control, and shipping are some of the other cost-producing procedures that need to be investigated at this point [30].

The creation of preliminary part drawings is covered in stage four; preliminary part drawings aim to assist the designer in focusing on their initial idea and ensuring that they are consistent with end-user requirements. In order to perform stress analysis and begin the detailed empirical design, preliminary drawings are strictly required. Empirical design is based on the designer's understanding of reinforced composite materials and processes that have been employed in similar situations. Estimates are taken concerning element size, exterior polish, and other traits based on experience with similar components and all other relevant information. The two useful technologies utilized in the examination of preliminary part model design choices and approaches are regarded as Finite Element Analysis and Computer Aided Design [31–33]. After assessing various design routes and options, the material engineer can determine which approach best meet the functional requirements. Many elements must be discussed and settled before the design can be committed to production drawings. Consultation with a reputable custom mold maker at this point, as well as elsewhere in the design process is necessary for guidance on design details.

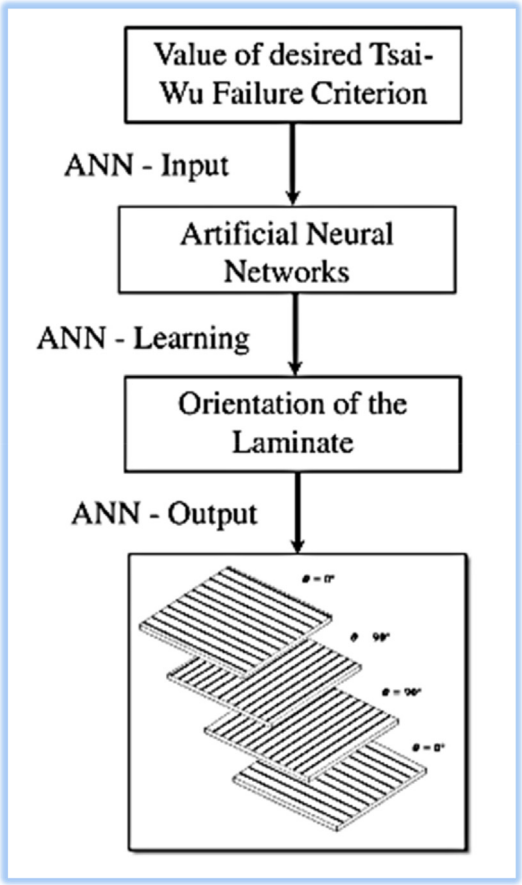


Fig. 5. Artificial Neural Network inversion technique for composite laminate layer optimization [87].

The factor screening and model simulation stage is characterized by the examination of a large number of control factors to identify the important characteristics that may have a significant impact on properties. The Plackett–Burman design of the experiment option allows the designer to evaluate a large number of variables in a small number of trials [34]. The optimal material factor combination is provided once the key variables have been discovered. The best material combination will be one that balances the cost of each material with the yield it produces. A functional prototype will be required to bring the performance of the optimal material factor mix to reality. Prototypes for applications using contact molding methods are the easiest to make. Prototypes of this type may easily be compared to performance and dimensional specifications [35]. During tool try-out runs, it's vital to keep temperature, pressure, and cycle lengths under tight control so that the parts produced can be evaluated against end-use goals. This level of control is also imperative to ensure that components from manufacturing runs are repeatable [36].

4. The place of data-driven material design and drawbacks of traditional methods of reinforced composite development

Notably, experimentation and high-throughput simulation are two regular strategies that are generally embraced in the field of materials science and design. Nonetheless, it is difficult to utilize these two strategies to speed up materials design and development due to the intrinsic limits of both exploratory conditions and hypothetical standing. As a rule, experimentation, which generally incorporates microstructure and property examination, property estimation, manufactured tests, etc, is a simple and instinctive strategy for materials research, even though it is time-consuming [37]. However, the success of experimental measurement of hybrid polymer composite materials depends on the quality of test facilities, the exploratory climate, and the designer's experience.

Furthermore, experimental study of hybrid composite material may be costly and time-consuming, especially when many trials are obligatory to examine each material variable. Also, there might be some flaws in the research design or procedures, as well as uncontrollable or unpredictable mistakes that may have a substantial influence on the study's findings. These issues arise because experimental material research often demands a precise degree of factor control and human error is a distinct possibility at any point during the investigation [38]. Therefore, in order to do appropriate experimental study of hybrid composite material, the study must isolate each variable and examine it separately; this is however a lengthy operation that requires major human and financial resources. Even if the factors are controlled, the result may provide internal validity to the detriment of outward validity. In any case, experimental material research has both advantages and downsides, indicating that it is a convenient system to utilize, but it must be strictly regulated to be

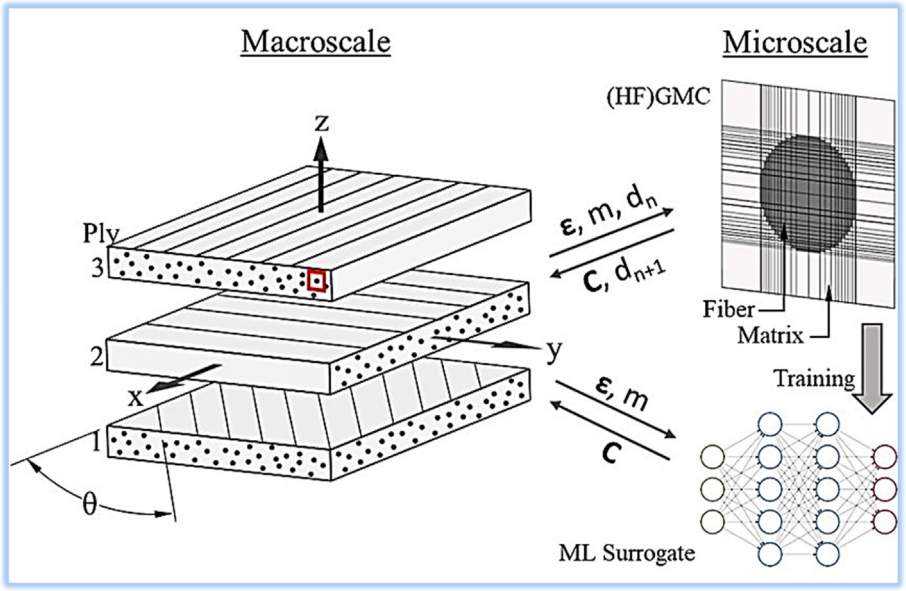


Fig. 6. ML **model for composite laminate micromechanics** [90].

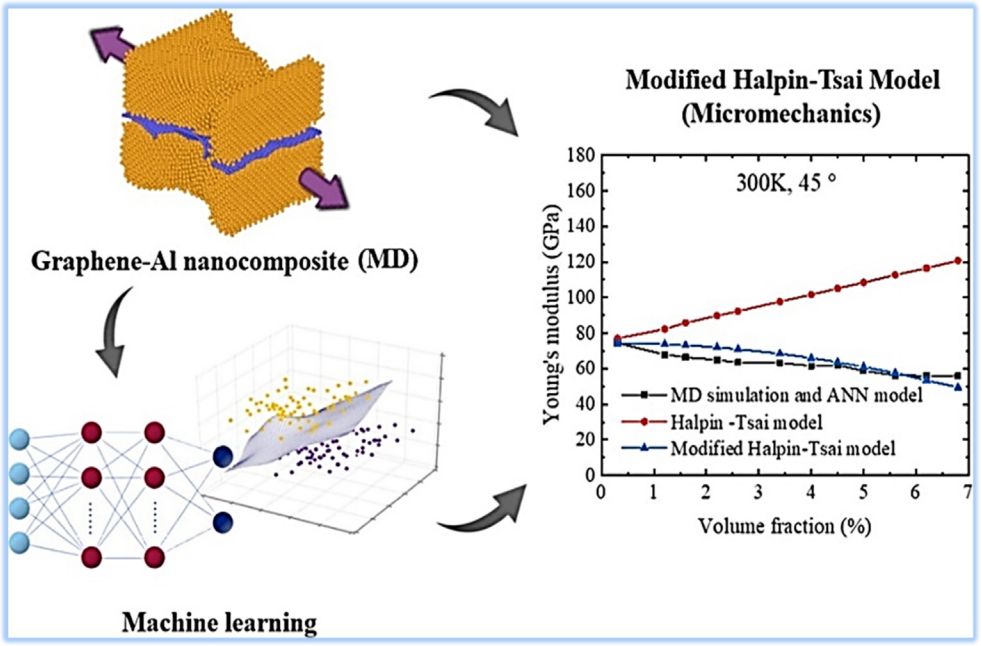


Fig. 7. Estimation of mechanical characteristics of graphene/aluminium nanocomposite via ML [92].

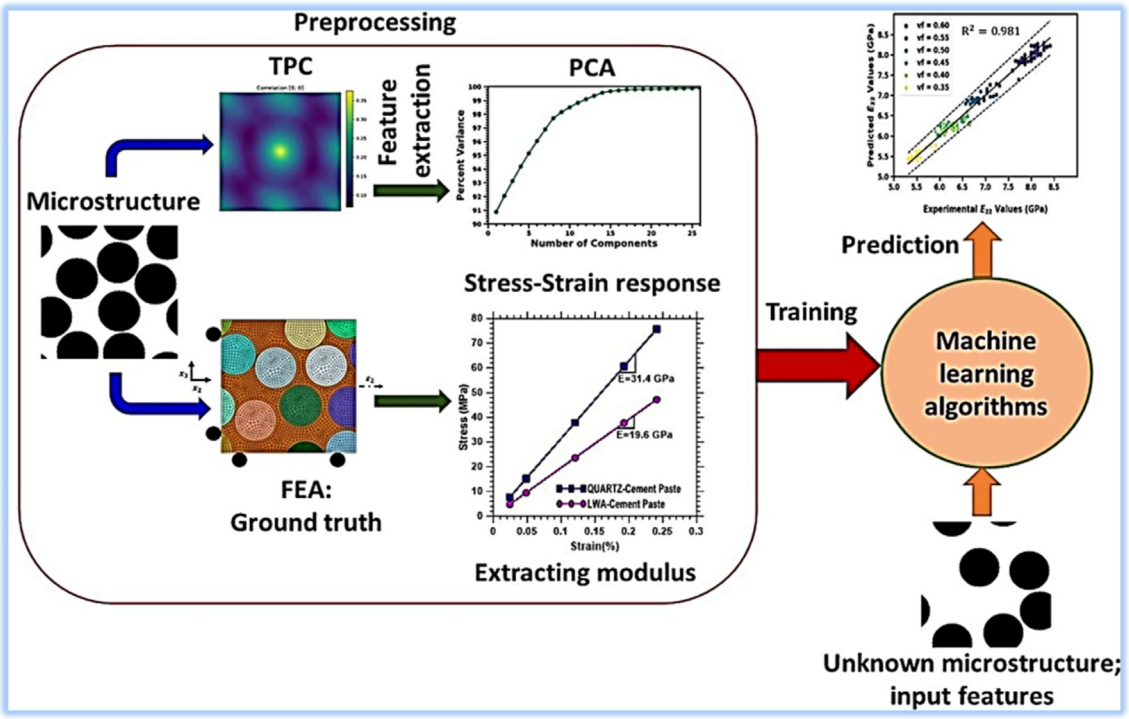


Fig. 8. Machine learning-based enhanced property forecast of composite materials properties [94].

effective [39]. Thus, composite material researchers must establish a balance between the benefits it provides and the downsides it causes by comparing and validating it with other approaches such as high-throughput simulation.

High-throughput simulation when compared with experimentation requires less time and is beneficial for providing genuine analyses in that the material designer has full power over the significant control factors [40]. Nevertheless, there are likewise numerous difficulties identified with high-throughput simulation; e.g., it unequivocally relies upon the microstructures of the materials in question; it is predicated on the idea that the researchers have reasonable estimates for material properties and how these depend on processing; it requires superior computer hardware and no express use can be made of past estimation results when a new framework is contemplated. In these cases, simulation is not useful by itself. Besides that, there are divergent statistical, computational, and mathematical methods for tackling material design problems which in turn raise fundamental issues surrounding multi-objective optimization [41]. In most cases, there are many limitations arising from the selection of control factors and their levels. Consequently, ML might provide an alternative route to data-driven reinforced composite design and development. Fig. 2

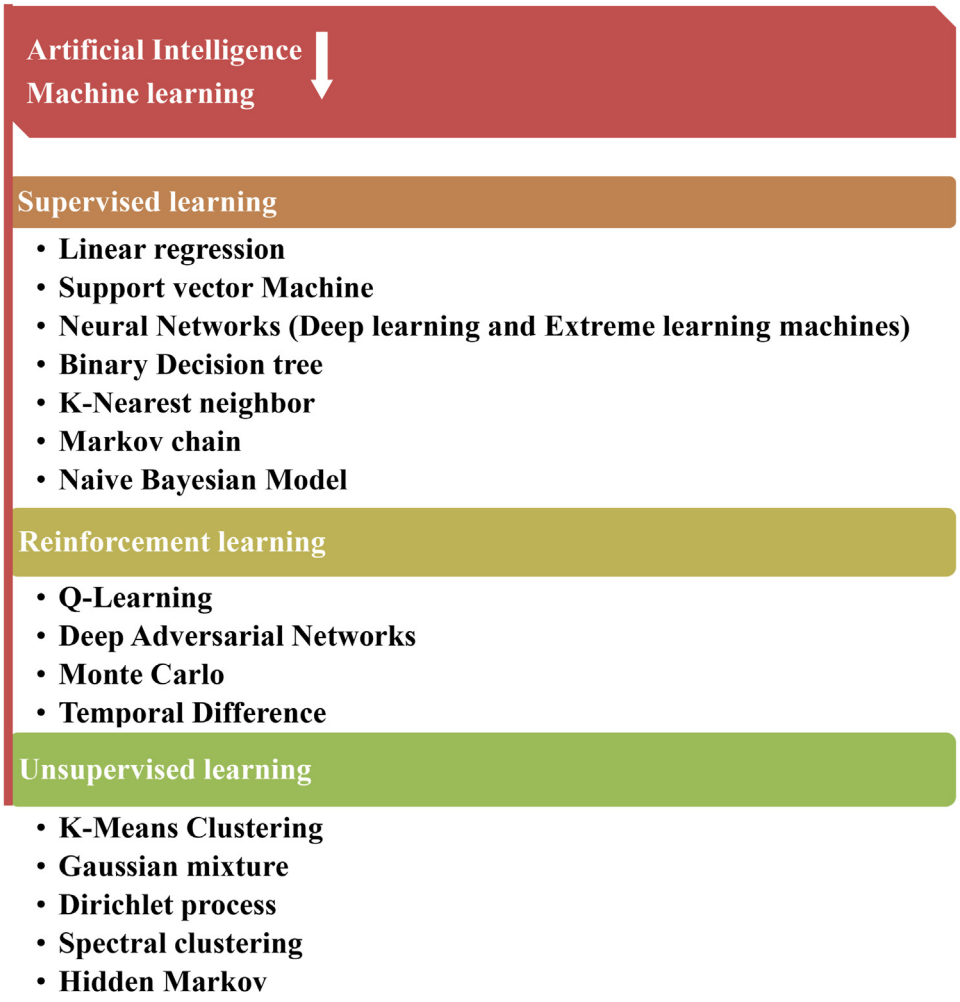


Fig. 9. Classification of ML

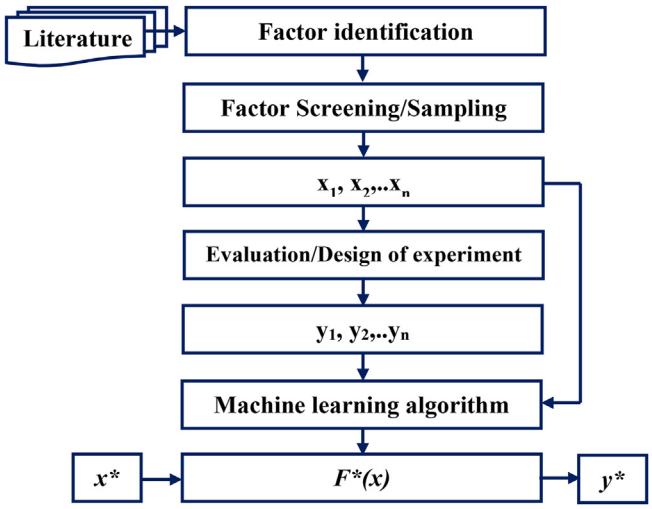


Fig. 10. System diagram of supervised learning framework.

shows a framework of data-driven design of composite material showing production routes, data acquisition sources and simulation tools. The given data-driven approach demonstrates predictive modeling and design decision support methodologies for a reinforced composite design using data from the manufacturing process, micromechanics, finite element method, literature, and electronic databases etc.

Micromechanics is a method that is frequently used to simulate reinforced polymer composites and other mechanical properties of materials at the microscale, and it is a source of data that can be used to make predictions using ML. The goal of composite micromechanics is to determine the effective amounts of the composite utilizing the materials that make up the composite, since this gives the fundamental information required for machine learning-aided design of reinforced polymer composites. Recently, Micromechanics technique has been deployed in literature for failure modeling of triaxially loaded unidirectional carbon fiber reinforced polymer composites [42], mechanical characteristics of composites with fiber reinforcement [43,44], model of viscoelasticity for shape-memory polymer composites [45] and Multi-phase degradation of moisture-exposed bio-based wood polymer composites [46].

The finite element method is a well-liked technique for the design and analysis of composites. The analysis of composites under various loading scenarios, such as quasi-static loading and dynamic loading is also done using finite element analysis. Sun et al. [47], conducted forward shape-change prediction via a recurrent neural network-based ML model using a training dataset obtained from finite element simulations. Their approach was considered very effective in designing active composites that can be produced in 4D. Recently, finite element method has been deployed in literature to gather data on the fracture behavior of polymer composites [48], behavior of cementitious cellular composites under auxetic conditions [49], physical characteristics of Cu–Al_2O_3 nanocomposites [50] and forecast of the fatigue life of adhesively composite material [51].

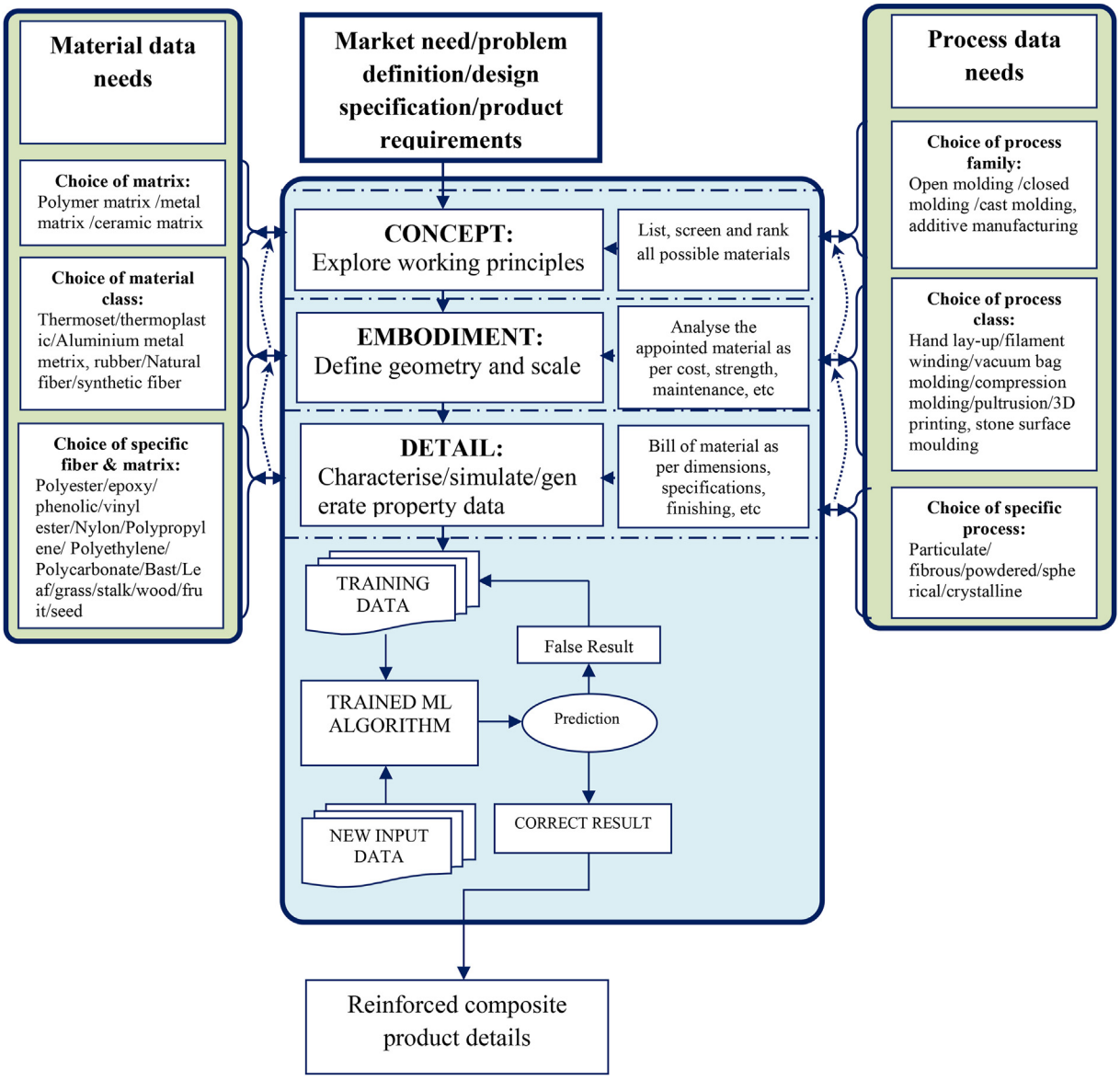


Fig. 11. The place of ML in reinforced composite material design and selection.

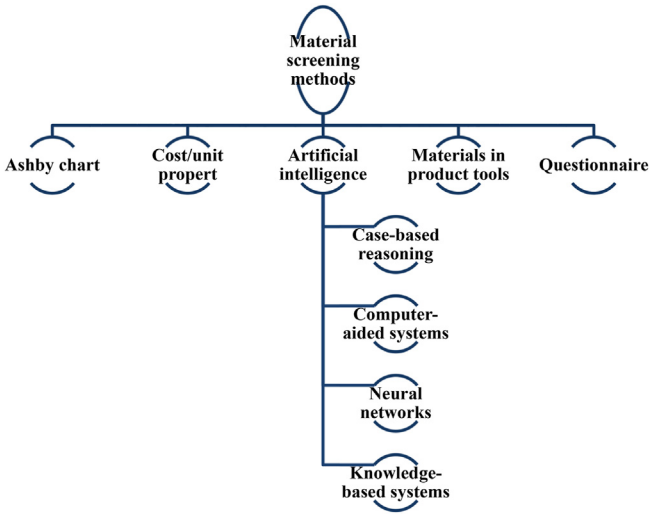


Fig. 12. The place of ML in material screening.

Recent data-driven techniques include everything from simple statistics derived from historical data to ML prediction models. By minimizing manual analysis and physical testing, data-driven reinforced composite design may typically result in greater engineering efficiency [52]. The capacity of various ML algorithms to examine enormous amounts of material data and find unseen trends and patterns without human involvement has led to widespread application in reinforced composite research. Multi-dimensional and multi-variety material property data are also well-suited to ML methods.

5. ML protocol in the technology of composite materials

ML system is built in three main phases involving data preparation, model creation, and model evaluation as shown in Fig. 3. According to Pilania [53], the first phase is data preparation, which involves gathering original data from computational simulations and experimental observations. The second phase is model creation, which includes connecting input and output data using a set of nonlinear or linear functions. ML model building contains three steps i.e., “ML technique selection”, “training data selection”, and “model building”. Complex interactions between the conditional elements and the goal qualities are common in

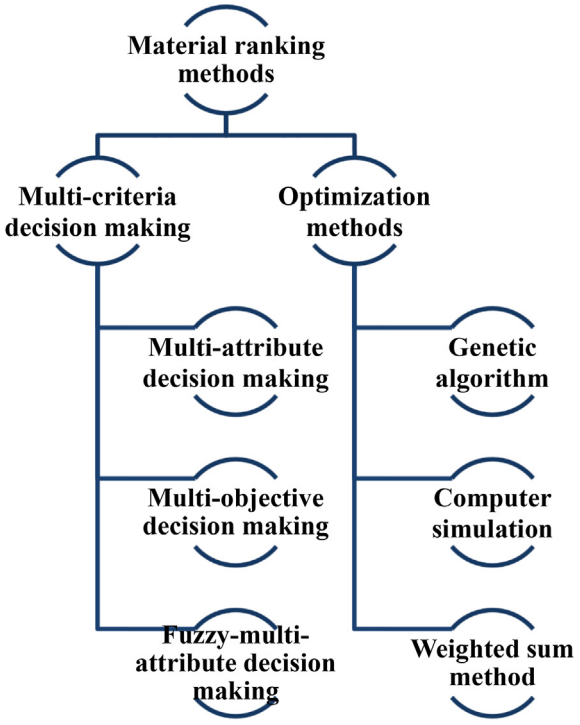


Fig. 13. The place of ML in material ranking.

materials science research, which standard approaches struggle to handle. Stach et al. [54], discussed various particular possibilities for materials science research and subsequently framed future investments in materials science and technology, software infrastructure and artificial intelligence.

ML as a pathway to artificial intelligence, has been shown to be useful in experimental domains with sparse data by Gongora et al., [55]. ML is interested in finding the coefficients with which a mapping function approximates a target function as closely as feasible using samples of the target function. The last phase is model evaluation, in which the designer engages calculation-based tests to assess the generalization mistakes of models and uses the findings to choose the best one. In the model evaluation phase, the predictive performance of the model selected algorithms, and explanatory variables on the predicted behavior are employed. For this purpose, testing data are required to assess a model's discriminative skills on a new dataset; the test error obtained on the testing data may then be used to approximate the generalization error.

ML and evolutionary computation are data-driven techniques, whereas fuzzy logic and probabilistic reasoning are knowledge-driven reasoning techniques [56]. Today, the domain of ML has been stretched out to incorporate a multitude of knowledge and new material discovery [57]. This is contrary to a traditional methodology that depended intensely on synthetic instinct consummated by difficult experimentation-based improvement cycles for novel materials disclosure; the conventional way to deal with materials development requires integrating and testing thousands of material options. Assuredly, the data-driven ML approach offers a significantly more productive and designated way to handle the feedback loop encapsulated in the new material development process [58]. Utilizing different ML algorithms, a material designer can essentially screen possible candidate materials, empowering quicker development in materials development and discovery.

ML offers a good way to use available datasets and construe information-driven models that can be utilized to speed up the development of new composite materials [59,60].

As an outstanding achievement, ML strategies have been utilized to speed up the four-dimensional printing design of dynamic composites [12], Alumina Reinforced Aluminium Alloy Composites [9], carbon nanotube-reinforced cement composites [61] and fiber-reinforced polymer composite strain prediction model [10]. Fig. 4 shows a generalized protocol for utilization of ML in reinforced composite design. For a start, certain information is produced to generate a data set of properties. ML is then used to surmise a few examples inside the dataset and build up a predictive property model for the composite [62–64]. ML algorithms can achieve three kinds of undertakings, such as supervised, unsupervised, and reinforcement learning.

6. Implementation of ML algorithms in reinforced composite material design

Materials design requires a knowledge of the cause-effect link between material structure and properties. The structure-property linkages can be thought of as "input-output" relationships that aid in the design of materials engineering systems. Hickman [65] described the relationship between the machine learning-aided design of reinforced composite lamina and resultant material properties. His conceptual framework was predicated on the traditional two distinct phases and bulk property's philosophy of composites. As influenced by specific material application, the material designer may target a certain factor level (say fiber spacing f_s and fiber diameter f_d) such that the desired material property (e.g., elastic modulus E_c) is maximized, alternatively, the material designer may select a reinforcement at an elastic modulus E_c for fixed particle size and/or diameter. The described design scenario is a constrained optimization problem in Eq. (1):

$$\left.\begin{array}{l} maximize\ E_c = F(m_E, f_E, m_v, f_v, f_s, f_d) \\ subject\ to\ m_E, f_E \in E \\ m_v, f_v \in V \\ f_s \in S \\ f_d \in D \end{array}\right\} \quad 1$$

where S, D, E, and V are sets of feasible design values for each parameter. Reinforced composite design task often requires full-scale analysis of the material's response to external conditions such as load, temperature, or pressure and evaluation of the structural reliability of the material before deployment in specific applications [66–68].

Within the last half-decade, machine learning-based analysis of reinforced composite response to external forces/effects for a range of design applications has blossomed [15,69–71]. In these analyses, there exist an assumed relationship between macroscopic composing factors (control factors) and measurable material properties (responses/target). Additionally, control factors may be discrete or continuous, depending on the nature of the parameter being constrained [72]. Factor specification and identification is the first fundamental task in ML initialization because it contributes to reducing the design space into a finite number of dimensions [41,73,74]. This is because, a large number of dimensions create a complex learning problem while few dimensions limit the accuracy of the model. Evidently, principal component analysis [75,76] and screening design [77] can be utilized to analyze the sensitivity of a ML model to changes in control factor levels.

Eq. (1) can be generalized by designating x-number of factors as a row vector $x = [x_1, x_2, ..., x_{d-1}, x_d]$. Every part of x addresses the worth of the factor $x_j, j = 1...d$. It is accepted that the objective function y is reliant upon the position of x in the design space. y is a multi-dimensional dependent variable estimated by tests. Meanwhile, the most constrained optimization problem can be generalized as

$$\left.\begin{array}{l} maximize\ y = F(x) \\ subject\ to\ x_j \geq a_i j = 1..d \\ x_j \geq b_i j = 1..d \end{array}\right\} \quad 2$$

Here (a_j, b_j) is the scope of achievable design for the control factor x_j. The achievable locale is characterized freely in each component of the control factor space. On account of adjusted or occasional material microstruc-

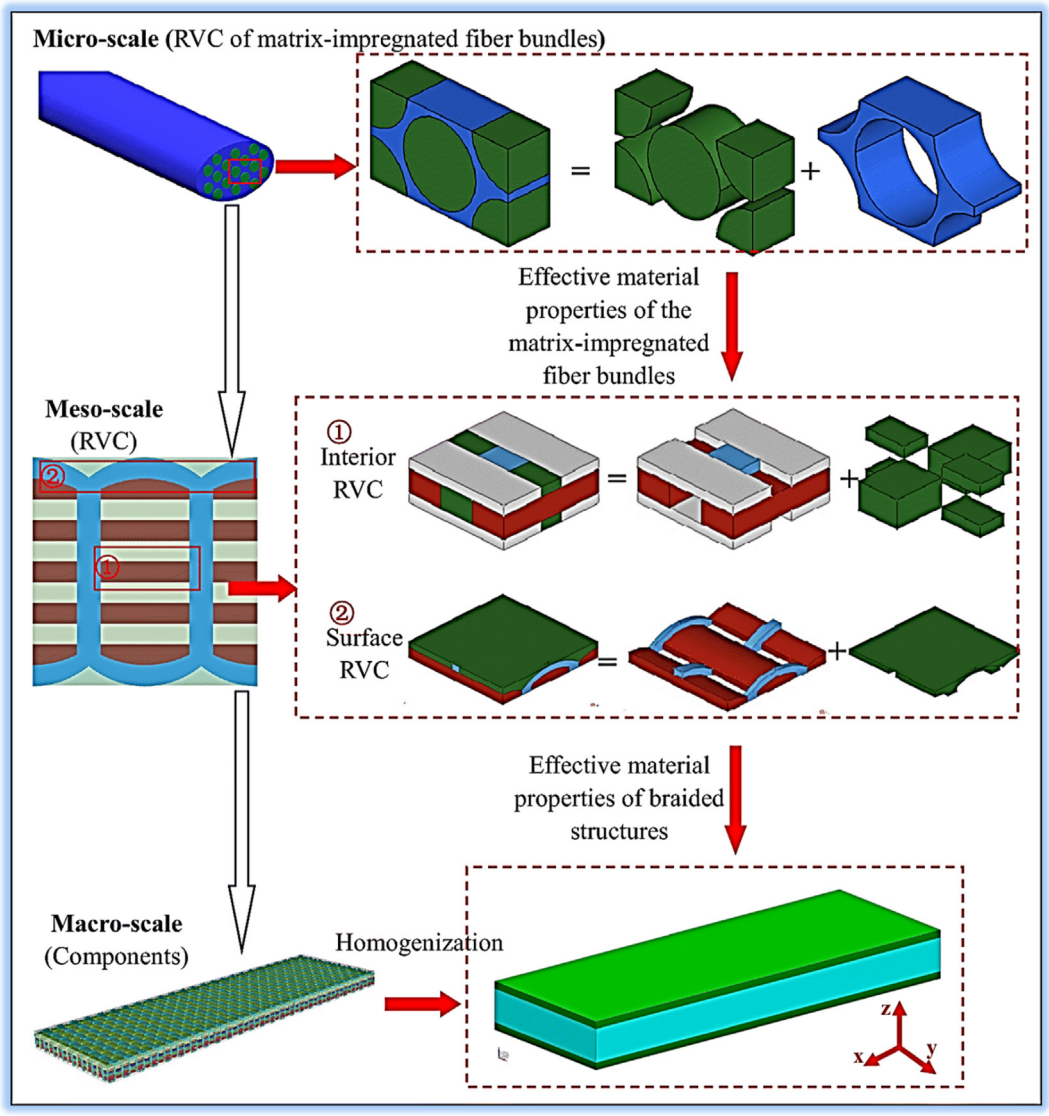


Fig. 14. FEA Modeling schema for reinforced composites at multiple scales [121].

tures, control factors like aspect ratio, fiber orientation, stacking sequence [77], volume fraction, particle size, curing angle, fiber curing temperature and curing time [5,6,78,79], fiber weight fraction [80] and fiber treatment concentration [81–83] are mostly considered as they strictly influence the properties of the composite lamina.

Although most measurable material properties can be determined using mechanics of materials principles. Mechanics of materials is limited to linear elastic systems [84–86]; hence the need for data-driven algorithms that can handle nonlinear behavior and damage in reinforced composites. Utilizing ML calculations, it is feasible to capture complex nonlinear characteristics within a large data set from tests. This will be useful to the analysts to consider the impact of control factors in an exceptional way. ML has been utilized to analyze laminated composite materials with high strength-to-weight and stiffness-to-weight ratios [73].

A laminated composite is made up of layers of fiber lamina composite materials that may be linked together to offer technical features like as in-plane stiffness, thermal expansion coefficient, bending stiffness, and strength. The laminate may display coupling between in-plane and out-of-plane responses depending on the stacking sequence, thickness, orientation, number of layers, and material characteristics of the individual lamina. Hereby, the optimization procedure is well suited for usage with a finite element program and a ML technique. Based on the Tsai–Wu failure criteria, Diniz et al. [87], suggested an artificial neural network inversion approach for efficient optimization of the layers of composite laminate (Fig. 5). Their research shows that neural networks may be used to investigate composite materials. Furthermore, their research showed that the composite laminate layup produced effective and consistent results that met the specified safety factor.

Techniques for the implementation of ML techniques in the prediction of statistical design of composite laminates were provided by Ref. [7]; their work addressed the groundwork for non-linear finite element simulations to forecast first-ply failure, ultimate strength, and failure mode of composite specimens, reducing the computing time necessary to practically acquire the design allowable for composite laminates. With the damage progression of various failure modes [88] used deep learning to monitor the performance degradation process and predict the residual load-bearing capacities of glass fiber/epoxy composite laminates. The authors devised a method for quantifying the degradation process and composite material damage tolerance. In a related study by Khan et al. [89], identification, quantification, and localization of damage to smart composite structures using ML approaches were further discussed. The representational capacity of the discriminative features for the relevant problems determines how well the ML paradigm for damage assessment

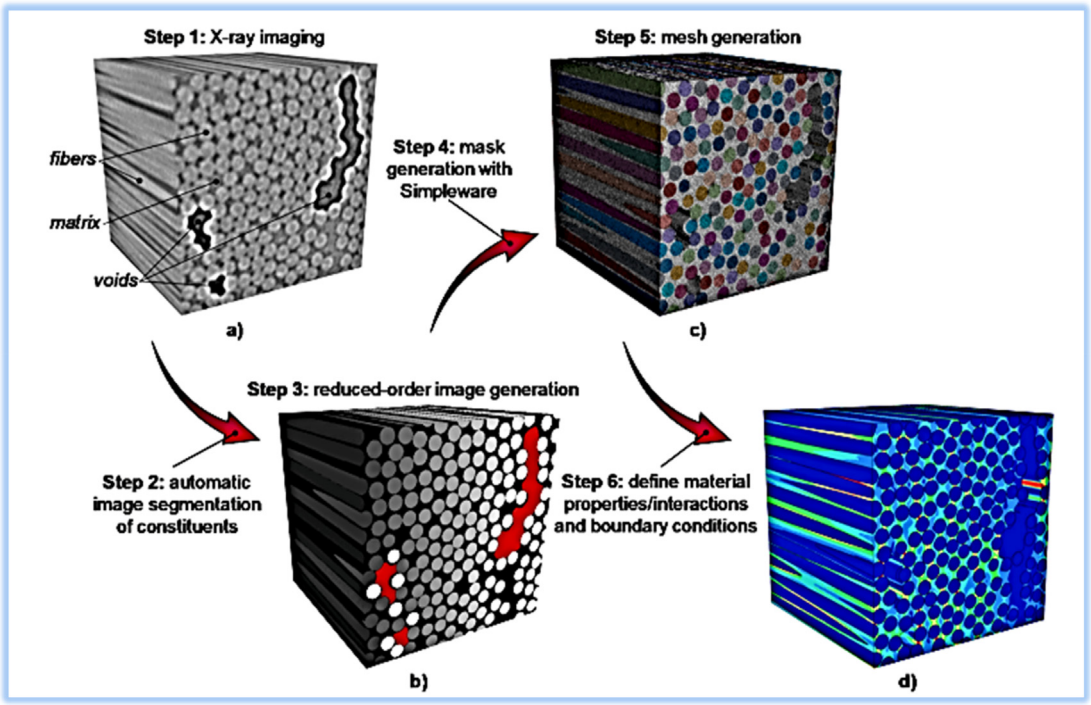


Fig. 15. The steps involved in converting tomographic picture data into a finite element mesh [125].

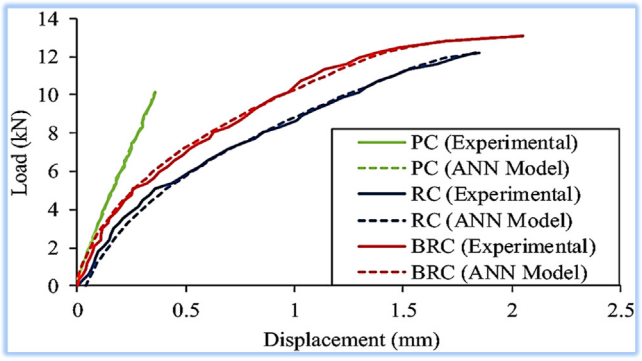


Fig. 16. Comparing the load vs deflection curve derived with the ANN model and experimental data [133].

performs. However, from a practical sense, it is impossible to construct a global or superset of discriminative features that might distinguish between the damaged and undamaged states of the structures while also highlighting different modes of failure.

Stuckner et al. [90], developed a computationally efficient ML surrogate model for micromechanics of composite laminates, as shown in Fig. 6; their surrogate model was able to produce a latent material damage representation that captured loading history and route dependency thanks to the use of long short-term memory layers in the neural network design. To detect optimum material qualities, ML calculations can be integrated with structure optimization [38]. ML has gradually become a highly fascinating tool in the overall reinforced composite research community, where numerous statistical and probabilistic approaches have been demonstrated to speed up both basic and practical research. The enormous success of ML algorithms in other sectors must have prompted material designers to investigate the idea of using them in the development of novel composites with improved characteristics and broader uses.

Modern ML algorithms excel at a wide range of tasks; however, it is

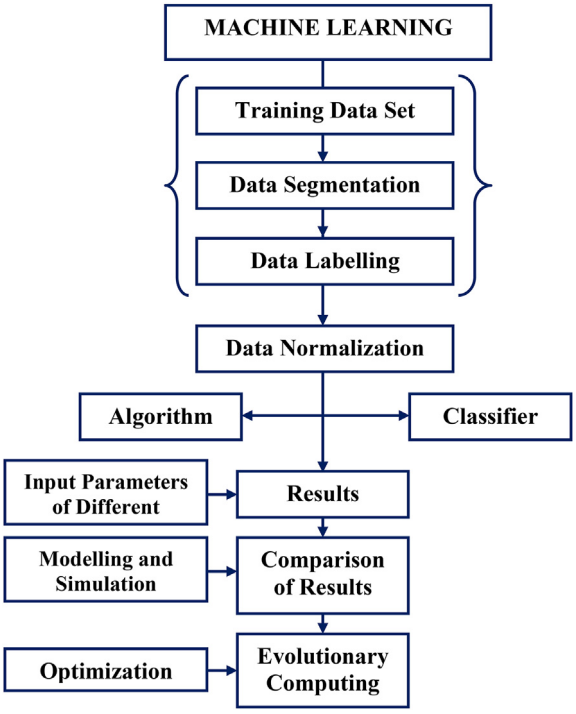


Fig. 17. ML execution process framework.

also well understood that these complicated models must deal with uncertainty arising from a variety of sources. As a result, determining whether the model is producing correct predictions or simply guessing at random is difficult, and assessing the confidence boundaries is critical. The uncertainty in Neural Networks used to predict fatigue in composite materials was estimated by Fernández et al., [91]. According to the authors, evaluating the uncertainty of ML models could lead to a reduction in safety factors in engineering challenges, as well as allow for

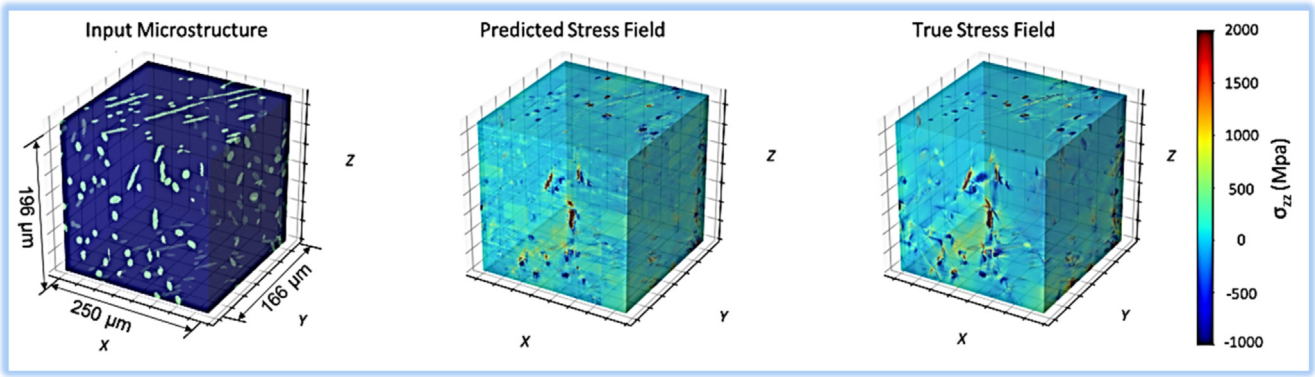


Fig. 18. 3D depiction of input microstructure, CNN predicted stress fields and the true stress field [150].

well-informed decisions in scenarios with a high degree of uncertainty and risk. In estimating the mechanical characteristics of graphene-reinforced metal matrix nanocomposites, Liu et al. [92], employed ML to avoid the high computing costs, scalability issues, and mistakes encountered with micromechanics-based reinforced composite models. The micromechanics-based Halpin-Tsai model was modified by integrating ML as illustrated in Fig. 7, permitting Young's modulus to be ascertained with markedly better certainty from a clear-cut correlation that is useful in the design of the nano-composite structure.

Microstructural characterization and control have grown more critical as reinforced composite materials are increasingly recognized for usage in sensitive practices. This is owing to accumulating body of literature that the configuration of fibers and microstructural properties of composite materials may be a major element in their performance [93].

Additionally, the rapid definition of spatial microstructural parameters and machine learning-assisted image analysis has been realized possible by recent advancements in image analysis and assessment software. Using microstructural images, Ford et al. [94], developed a ML system (Fig. 8) to predict the mechanical characteristics of complicated, multi-phase composites. Microstructural science's core tool is the quantitative representation of microstructure, which connects the structure of the material to its composition, process history, and attributes. Traditionally, microstructural quantification entails a human selecting what to measure a priori and then inventing a mechanism to do so. Recent breakthroughs in data science, such as computer vision and ML, have opened up new possibilities for extracting information from microstructural pictures

7. General classification of ML algorithms

ML is split into three groups according to the learning technique utilized in the system's training as displayed in Fig. 9. There are three types of learning methods: supervised, unsupervised, and reinforcement learning [95]. In the supervised learning method, the system is first put up as a model with the ability to learn but is underdeveloped or 'ignorant' until it is taught. The training procedure entails giving the model a collection of input and output data, such as x and y, for which an association or link must be established. During training, supervision refers to the presentation of the proper output data that matches the input data to the model. As previously said, the training phase entails repeated or adaptive adjustments to the model until it has learned to correctly associate or build a link between the input and output data to a fair degree of accuracy. Once the model has been trained, it is tested on a set of input and output data that has never been seen before. The model's performance is measured by how well or correctly it predicts the outputs that correspond to the test inputs. The degree of nonlinearity in the interactions between the input and output data specified the accuracy that can be achieved.

The field of supervised learning can be isolated further into linear regression, support vector machine, neural networks, decision tree, nearest neighbor, Markov chain, and naive Bayesian model. In general, the supervised ML structure comprises factor identification through literature search [96,97], experience, experiment, factor screening/sampling to produce arbitrary input vectors x, followed by a framework that produces yields y for a given input vector x and finally, a learning machine that forecast obscure qualities y^* at test point x^* utilizing training data x, y. Fig. 10 outlines the supervised learning structure.

Reinforcement learning is a branch of ML concerned with executing appropriate actions to maximize reward in a given circumstance. It is all about finding out how to behave in a specific scenario such that the idea of cumulative reward is maximized. Reinforcement learning differs from supervised learning in that it does not need the display of labeled input/output pairs or the explicit correction of sub-optimal behavior. Instead, striking a balance between exploration and exploitation should be the top objective. Reinforcement learning varies from supervised learning in that supervised learning contains an answer key, allowing the model to be trained with the right answer, but reinforcement learning does not, and instead depends on the reinforcement agent to select how to finish the job. In the absence of a training dataset, it is bound to learn from its experience.

In unsupervised learning strategies, the system looks for structures inside the input without considering the output response y, while the supervised learning method search for model $F^*(x)$ that most precisely creates output response y from input information x. The ML model is solely provided with input data in unsupervised learning. During training, the model learns to organize the information provided into clusters. There is no doubt that the model requires direction, such as the number of groups that should be produced. The model is trained on a task by enabling it to execute random actions to achieve a goal in reinforcement learning. The model is given a bonus if it succeeds and a penalty if it fails. It learns the optimal approaches to meet the goals through this process across many iterations.

In terms of developing hard laws of correlations between input elements and parameters and product output, materials design, processing, and characterization modeling have been stringent. This is currently the case due to a lack of understanding of all interacting components and the difficulty of controlling processing conditions at various locations inside the material. As a consequence, there appears to be significant evidence that it is impossible to make a universal conclusion about which machine-learning algorithm is the best [98]. Different ML algorithms for identifying delamination in composite beams were examined by He et al., [99]. The authors observed that the support vector machine gained the best prediction performance for fiber-reinforced polymer composite beams in comparison to back propagation neural networks and extreme learning ML approaches, with superior predictive efficiency and just minimal sample sizes needed.

Table 2
Appraisal of MATLAB deployment in machine learning-based design of reinforced composite.

S/N	ALG	DS	Control factors	Target	Ref
1	ANN	FEA	Compressive strain, coating thickness, a fraction of coated sand	The fractional change in resistance	[153]
2	NBM	SIM	Feed rate, kill rate, diffusion coefficients	Damage Tolerance	[154]
3	ANN	EXP	Slurry Pressure, Impingement Angle, Nozzle Diameter	Erosion rate	[155]
4	DNN	EXP	Hybridization layer	Tensile strength, modulus, failure strain, shear strength	[156]
5	ANN	EXP	Electrolyte concentration, interelectrode gap, duty factor, voltage	Material removal rate	[157]
6	ANN, SVM, ELM	SIM	Modal frequencies	Interface delamination parameters	[99]
7	ANN	EXP	Volume percentage, load, sliding velocity, sliding distance	Wear rate, S/N ratio	[151]
8	SVR, BDT, GPR, RF	FEA	Fingerprints	Compression stresses, Mass	[158]
9	ANN	EXP	Fibertype, electrode distance, max. Absorbed energy, max. Impact force, initial resistance	Damage localization in x-y coordinate	[159]
10	LN, RT, SVM, EBT, FL,ANN	LIT	Density, average length, average diameter, weight fraction, CNT surface modification technique, Young's modulus of resin, the tensile strength of resin, density, average length, average diameter, weight fraction	Tensile strength	[96]
11	ANN	FEA	Thermal conductivity of the fibers and the matrix, the Kapitsa resistance, volume fraction, aspect ratio	Macroscopic thermal conductivity of the composite	[160]
12	KMC	Stack of images	Set of fiber positions	Chirality of the fibers	[161]
13	ANN, SVM, ABR	SIM	The volume fraction of graphene, alignment angle, chirality, and temperature of the surroundings	Young's modulus and ultimate tensile strength	[92]
14	ANFIS, ANN	EXP	particle size, curing duration, power level, volume percentage, and curing angle	tensile strength	[79]
15	ANN	FEA, EXP	Friction effect, bolt-hole clearances, tightening torques	Minimize uneven load distribution	[119]
16	MPMR, MARS	FEA	Poisson's ratio, ply-orientation angle, stacking sequence, skew angle, thickness, mass density, elastic modulus	Buckling strength	[162]
17	ANN	EXP	Density, temperatures	Flexural strength	[163]
18	XGBoost, MOR, CM	FEA, EXP	Force–displacement curves	Traction–separation relations at interfaces	[164]
19	GP	FEA	Braid angle and number of layers	Minimum mass	[165]
20	ANN	FEA	Strength values of center-cracked specimens	crack resistance curves	[119]
21	LR, RR, ABR, BR, ET, GB, RF	FEA	Eigenvectors and eigenvalues of the diffusion tensor	constitutive model parameters	[166]
22	ANN	FEA	Mesoscale architectural void features	Poisson's ratio	[49]
23	ANN	WWFE	Longitudinal stress, Transverse stress,	Safety Status: Failed/Survived	[167]
24	ANN	EXP	The torque and the slamming load	Temperature trends	[168]
25	KNN	EXP	Aspect ratio, time, volume fraction, elastic modulus	Stress Intensity Factor	[169]
26	ANN	EXP	Deposition Rate, hot gas torch temperature, Nip-point Temperature, Consolidation Force	Elastic Modulus, Short-beam strength, Flexural Stress, Flexural Strain	[62]
27	ANN	EXP	Concentrations of Titanium dioxide and cellulose nanocrystals	Fracture strength, yield strength, Young's modulus	[170]
28	ANN	EXP	laser power, welding current, defocused distance, and welding speed	welding joint morphology	[132]
29	ANN	EXP	Stress Level, Applied Bending Stress	Fatigue life	[171]
30	ANN	EXP	Infiltration temperature, Infiltration pressure, Density	Bending strength, Hardness	[172]
31	ANN	CHS	fiber positions and volume fraction	Effective elastic modulus	[171]
32	ANN	EXPERIMENT	Fiber length, Biomass amount, Time	Water absorption	[173]
33	ELM, GRNN	EXP	Geometry dimension, stress, and orientations	Fatigue cycle	[57]
34	ANN	EXP	Pressure, temperature, impregnation distance, viscosity	Impact energy absorption	[174]

KEY.
SIM=Simulation, EXP = Experiment, ANN = Artificial Neural Network, ALG = Algorithm, FEA=Finite Element Analysis, NBM=Naive Bayesian Model, SVM= Support vector machine, ELM = Extreme learning machine, SVR=Support Vector Regression, GPR = Gaussian Process Regression, RF = random forest, BDT=Binary Decision tree, LN = Linear Regression, RT = Regression Tree, EBT = Ensemble Boosted Tree, FL= Fuzzy Logic, KMC=K-Means Clustering, ABR = AdaBoost Regression, ANFIS = Adaptive Network-based Fuzzy Inference System, MPMR = Minimax Probability Machine Regression, MARS = Multivariate Adaptive Regression Spline, XGBoost = eXtreme Gradient Boosting, MOR = Multi-output Regressions, CM=Classifier Models, GB = Gradient Boosting, GP = Genetic Programming, LR = Lasso Regression, RR = Ridge Regression, BR= Bagging Regression, ET = Extra Trees, WWFE=World-Wide Failure Exercise, DS = Data source, LIT = Literature, KNN—K- Nearest Neighbor, CHS=Computational Homogenization Scheme, GRNN = Generalized Regression Neural Network.

8. Integration of ML in reinforced composite materials design and process selection

As presented in Fig. 11., materials data is necessitated at each stage of concept, embodiment, and detailed design stages, yet at totally different degrees of breadth and exactness. The dotted lines indicate the iterative idea of the initial design and the way to follow in design refinement. A comprehensive classification of reinforced composites based on matrix and reinforcement has been reported elsewhere [100,101]. In addition, detailed discussions on the choice of process class and composite production routes are contained in Rajak et al. [102], additionally, specific procedures for choice of specific process with respect to particulate fibre development are covered in literature [103–105]. The selection of materials is a vital piece of the reinforced composite design and improvement measure (Fig. 11). To make reinforced composite products that can play out their capacity in determining working conditions over their plan life, proper reinforcement and matrix materials should be utilized. Normally, the criteria for materials selection are characterized by the product function, objective, prevailing constraints, the material's ease of processing and part manufacturability, the shape of the product, reproducibility, and assembly.

Because of the enormous assortment of matrix, reinforcement, and related manufacturing processes (as captured in Fig. 11), the choice of materials for a given part might be a difficult undertaking. Doubtlessly, if the choice interaction is not thorough, the designer might decide on an improper material or neglect an appealing elective material. Therefore, the deployment of a data-driven material selection strategy during the design stage of reinforced composite products is considerably

Table 3
Appraisal of Tensor Flow deployment in machine learning-based design of reinforced composite.

S/N	ALG	DS	Control factors	Target (from the title)	Ref
1	CNN	FEA	Random microstructures	Toughness and strength	[8]
2	RNN	CT-scan image	True angle label	Orientation and Defects detection	[149]
3	ANN	FEA	Slopes of the Load-Pin Opening, Displacement curve, Maximum load, the ratio of deformation at a given load, deformation at the maximum load	Fracture energy, peak stress	[175]
4	GBR, ANN	EXP, SIM	Type of fiber, Fibre diameter, Embedded length, Young's modulus of fiber, Poisson's ratio of fiber, Type of matrix, Young's modulus of the matrix, Poisson's ratio of the matrix, Loading rate, Preparation temperature, Test temperature	Interfacial properties of fiber-reinforced composites (shear strength (IFSS) and maximum force (F_{max})	[176]
5	ANN	EXP	Initiation energy, Propagation energy, Total energy, Ductility index, Tensile strength, Modulus of elasticity, Percent elongation, Net weight of fiber	Filler content for cotton fiber/PVC composites	[136]
6	ANN, RNN	FEA	Strain, fiber angle, elastic modulus	Compressive damage in composite laminates	[177]
7	ANN, FE	Microstructural descriptors and FEA	Volume fraction	Elastic modulus, Poisson's ratio, and Phase stresses	[94]
8	ANN	EXP	Fiber percentage, characteristic load	load vs. displacement curves	[178]
9	ANN, micro-mechanics	FEA	theoretical data from the Tsai-Wu criteria	Failure prediction under triaxial loading	[42]
10	CNN	FEA	Modulus, toughness, volume fraction of base materials; topology (architecture)	Effective toughness, strength, stiffness	[179]
11	ANN	EXP	Crashworthiness parameters	load carrying and energy absorption capability	[180]
12	DNN	ND	Front velocity, maximum temperature, characteristics lengths	The time constant, activation energy, total enthalpy of reaction	[137]
13	CNN	3D Printing	Tomographic images	Fiber orientation, defect detection	[181]
14	DNN	FEA	Composite constitutive laws	Young's modulus of fiber in the x_2 direction	[182]
15	FFDLNN	LHS	MSG	Yarn failure criterion	[183]
16	ANN	FEA		Electro-mechanical response	[184]
17	CNN	μ-CT scan image	experimental CT scans of GFRP polyamide 66	fiber orientation and anisotropy	[185]
18	CNN	EXP, FEA	Microstructure	Stress distribution field	[150]
19	ANN	EXP	normalized specific energy, load	Filler content	[136]

KEY.
EXP = Experiment, ANN = Artificial Neural Network, ALG = Algorithm, FEA=Finite Element Analysis, RNN = Recurrent Neural Network, GBR = Gradient Boosting Regressor, SIM=Simulation, FE=Forest Ensemble, CNN=Convolutional Neural Network, DNN = Deep Neural Network, ND = Data Generated with Normal distribution, FFDLNN = feed-forward deep learning neural network, LHS = data generated using the Latin Hypercube sampling, MSG = microscale failure analysis based on the mechanics of structure genome, DS = Data source.

fundamental. Various ML tools and computer algorithms have been utilized in recent times for materials selection all of which depend on a huge information bank of materials and their properties [106].

The two key strides in materials selection are screening and ranking as captured in Fig. 12. Whilst screening empowers the designers to rapidly limit the assorted material options to a sensible few, ranking limit the decisions further and afterward assess and positions the decisions to recognize the ideal material [107–109]. Despite the idea of matrix and reinforcement material selection is not new in composite product development, there are just a few types of research that have focused on applying materials selection procedures to the composite design process.

The ideal material choice to satisfy given useful design needs generally necessitates that a trade-off is struck between conflicting criteria. Fig. 13 shows assorted methods recently deployed by researchers for efficient ranking in material selection, these methods include multi-attribute decision making [110], multi-objective decision making [111], fuzzy-multi-attribute decision making [112], genetic algorithm [113], computer simulation [114] and Weighted sum method [115]. Therefore, the selection of materials is of prime importance in the reinforced composite product design and manufacturing process, which can be a valuable decision-support approach for engineers to settle on a proper material decision [116,117]. Scholars have investigated the relevance of material selection protocols from various fields considering various necessities such as green assembling and economic advancement [118]. In any case, ML has proven its efficacy in enabling optimal material choices.

The literature has demonstrated that machine learning-aided composite design models are viable in calculating the optimum fiber materials and layup stacking orientations. Qiu et al. [119], have also developed a deep learning-based composite design approach for efficient material and layup order selection from databases. Using a conventional layup design guideline, the authors reported reasonable solutions for various design boundaries such as target strength, maximum deformation, minimum thickness, and lowest cost.

9. Reinforced composite data sourcing and processing for machine learning-based design

The relevance of data in the machine learning-based design of reinforced composite materials cannot be overstated. Materials scientists can gather data via searching public databases or by performing computations or tests on their own. In most cases, the materials science machine learning-based design technique involves supervised learning based on a significant amount of data. As a result, methods for obtaining large amounts of data, such as high-throughput computation or experimentation, are necessary. Materials databases are becoming the primary repository for storing the structures and characteristics of materials, because of the rapid development of Materials Genomes. In this sense, Song et al. [13], published a complete list of material databases that are currently accessible.

Developing a good ML model needs both top quality and enormous amounts of data. Many attempts have been made to increase repeatability and minimize inaccuracy in materials databases to solve this problem. The FAIR (Findability, Accessibility, Interoperability, and Reusability) dataset rules, which are crucial for database building, were explored by Ref. [120]. These recommendations will assist academics in building better databases and enhancing the precision of established models that have been trained on them. Meanwhile, undesirable data, such as results from failed studies that are often lost in databases or publications, might be productively augmented.

Table 4
Appraisal of Scikit-learn deployment in machine learning-based design of reinforced composite.

S/N	ALG	DS	Control factors	Target	Ref
1	CNN	FEA	Microstructures and stress-strain curves	Modulus, strength, toughness	[186]
2	GBR, SVR	FEA, TCF	Microstructures	Ballistic impact resistance	[74]
3	ANN, KNN, SVM GBM, RF	LIT	Graphite content, hardness, ductility, yield strength, silicon carbide content, tensile strength, processing procedure, heat treatment, normal load, sliding speed, and sliding distance	Coefficient of friction (COF) and wear-rate	[52]
4	DT KNN RF SVM	EXP, FEA	Crack position, pattern type, loading angle, sample thickness, crack length	Fracture behavior	[48]
5	SVR, ANN, GBRT, RNN)	EXP	Temperature, strain, time, and draw ratio	predicting tensile strength and tensile modulus	[187]
6	DTR, ABR	EXP, FEA	The weight fraction of DSP, EPDM, NC, PP, PSP, SLF	Fracture toughness	[188]
7	FCNN, DNN, RBF, ANN, SVR, KNN	FEA	Heating rates, holding times, Holding temperatures,	Uniformity of the Degree of Cure	[189]
8	ABR, BDT GBPR, RF, XGBoost	FEA, EXP	Part insert temperature, Melt temperature, Mold temperature, Packing pressure, and Flow rate	Interface bond strength quality	[127]
9	RF, GBR, GB, SVM, KNN, KRR	experiments	milling time (MT, Hours), dislocation density (DD, $1/m^2$), average particle size (PS,nm), density (gm/cm3), and yield stress (MPa)	Vickers Hardness (MPa)	[190]
10	DFNN	FEA	MKEV, TND	Damage detection	[191]
11	GBR	FEA	Fiber volume fraction, the diameter of the fibers, inter-fiber distance, fiber clustering, periodicity	Young's modulus, transverse shear modulus, transverse normal yield strengths,	[129]
12	CNN	FEA	Microstructures and reduced stress-strain	stress-strain response	[192]

KEY.
TCF = Two-point Correlation Function, SVR= Support Vector Regression, GBR = Gradient Boosting Regression, DT = Decision Tree, RF = Random Forest, KNN=K-Nearest Neighbors, EXP = Experiment, ANN = Artificial Neural Network, ALG = Algorithm, FEA=Finite Element Analysis, CNN=Convolutional Neural Network, DS = Data source, SVM= Support vector machine, GBM = Gradient Boosting Machine, LIT = Literature, GBRT = Gradient Boosted Regression Trees, RNN = Recurrent Neural Network, DSP = Date Seed Powder; EPDM = Ethylene–Propylene–Diene Terpolymer; NC=Nano-Clay; PP=PolyPropylene; PSP=Pistachio Shell Powder; SLF=Short Latania Natural Fibre, DTR = Decision tree regressor, ABR = Adaptive boosting regressor, FCNN=Fully Connected Neural Network, DNN = Deep Neural Network, RBF = Radial Basis Function, ABR = AdaBoost Regression, BDT=Binary Decision tree, GB = Gradient Boosting, XGBoost = eXtreme Gradient Boosting, PR=Polynomial Regression, KRR=Kernel Ridge Regression, DFNN = Deep Feed-forward Neural Network, MKEV = Elemental modal kinetic energy values, TND = Translational Nodal Displacements.

Table 5
Appraisal of Weka deployment in machine learning-based design of reinforced composite.

S/N	ALG	DS	Control factors	Target (from the title)	Ref
1	Not stated	X-ray μCT images	changes in the percent, distribution, and connectivity of pores	Pore evolution, porosity percentage, and shape of shrinkage cracks	[193]
2	Not stated	XCTV	Micro-crack formation and propagation mechanisms	Crack segmentation, damage shape, size, spatial distribution, and density	[194]
3	CNN	FEA, XCTV	Segmented microstructure	Stress field	[150]
4	RF	EXP	Flow rates, volume slices	Void content, pixel-based segmentation.	[195]
5	KFE	FEA	X-ray microcomputed tomography (μCT) image	interactions between constituents, contact/friction between fibers, and interactions at the fiber-matrix interfaces	[193]
6	ZR, NBM, MLP, SVM, AB, DT	FEA	Damage sensitive features	Delamination damages	[196]
7	Not stated	Grayscale μ-CT images	Fiber volume fraction, fiber orientation distribution, and fiber length distribution	Porosity volume fraction, orientation tensor	[197]
8	Not stated	EXP, SIM	Hydrostatic stresses	Macroscopic tension, location of micro-void nucleation	[198]
9	ANN, MLP, SVM	EXP	Acoustic emission signatures	Fiber breakage, matrix cracking, delamination, and debonding	[199]

KEY.
EXP = Experiment, ANN = Artificial Neural Network, ALG = Algorithm, FEA=Finite Element Analysis, CNN=Convolutional Neural Network, DS = Data source, SVM= Support vector machine, XCTV = X-ray Computed Tomography Volumes, RF = Random Forest, KFE=Kalman-Filter Estimation, ZR = ZeroR, NBM=Naive Bayesian Model, MLP = Multilayer Perceptron, SVM=Support Vector Machine, AB = Adaboost, DT = Decision Tree, SIM=Simulation.

Scientific community can gather data from existing databases or literature sources, or they can create their databases using experiments or high-throughput simulations. However, because machine learning-based design models require certain data formats (e.g., pictures, texts, and graphs) as input, datasets must be pre-processed before it is used for model training. To that effect, data augmentation strategies may be deployed to grow datasets during pre-processing, and unnecessary data points that might degrade the model's performance should be deleted.

Table 6
Appraisal of R-language deployment in machine learning-based design of reinforced composite.

S/N	ALG	DS	Control factors	Target (from the title)	Ref
1	KNN, SVM	EXP	Nano-indentation	Nano-hardness, maximum load	[200]
2	ANN, CT, SVM	EXP	Nano-indentation	interfacial reinforcement	[201]

KEY.
EXP = Experiment, DS = Data source, ANN = Artificial Neural Network, ALG = Algorithm, FEA=Finite Element Analysis, KNN = k-Nearest Neighbors, SVM=Support Vector Machine, CT=Classification Tree.

Table 7
Appraisal of RapidMiner deployment in machine learning-based design of reinforced composite.

S/N	ALG	DS	Control factors	Target (from the title)	Ref
1	RF	EXP, LIT	Diameter of the bar, the Volume fraction of fibers pH of the solution, Temperature, Time	Tensile strength	[141]

KEY.
EXP = Experiment, ALG = Algorithm, DS = Data source, LIT = Literature, RF = Random Forest.

Table 8
Appraisal of unspecified software deployment in machine learning-based design of reinforced composite.

S/N	ALG	DS	Control factors	Target (from the title)	Ref
1	ANN	FEA, EXP	Material, thickness, layup, and setup	Blank temperature predictions	[67]
2	FCNN, DNN, RBF, SVR, KNN	EXP	Heating rate, holding time, holding temperature holding-stage cure profile	Uniformity of the Degree of Cure	[186]
3	ANN	EXP	Impact velocity, Impingement Angle, RBD Content, Erodent Size, Erodent Temperatures	Erosion rate	[202]

KEY.
EXP = Experiment, ALG = Algorithm, DS = Data source, FEA=Finite Element Analysis, ANN = Artificial Neural Network, FCNN=Fully Connected Neural Network, DNN = Deep Neural Network, RBF = Radial Basis Function, a Support Vector Regression, KNN=K-nearest Neighbors, CNN=Convolutional Neural Network, NRM=Nonlinear Regression Model.

Data augmentation is a technique that can assist in enhancing the diversity of training data while also reducing over fitting. In terms of the quantity, if the training data volume is insufficient and hard to expand, an agreeable strategy is to perform data manipulations on the initial data, such as cutting and rotation, to produce new data.

9.1. Finite element analysis (FEA)/high-throughput simulations

Designers have more latitude to develop characteristics and regulate the number and distribution of datasets generated while doing high-throughput experimentation or computations. Designing structures with composite materials presents unique problems, owing to the necessity to design both the material and the structure at the same time. Nonetheless, commercially available Finite Element Analysis software such as ANSYS™, SimScale™, COMSOL Multiphysics™, OpenFoam™, ABAQUS™, Altair HyperWorks™, Autodesk CFD™, IVRESS™, Nastran™, SAMCEF™, and LS-DYNA™ can be used to calculate the elastic, mechanical, and thermal properties of reinforced composite materials. Engineers may use Finite Element Analysis software to improve microstructures and homogenize material characteristics in complicated materials and composites. Fig. 14 shows an FEA modeling schema for reinforced composites at several scales published by Yu et al. [121]. Material characteristics calculated with Finite Element Analysis software can be utilized also as input data in machine learning-based synthesis [122].

For effective machine learning-based synthesis/network training and validation, an adequate measure of data ought to be gathered. Without considering the expense factors, experimental characterization may not provide sufficient data for model training and checking. Thusly, research evidence has shown that a finite element model can be created and optimized to produce the required material property data [123]. Before the advent of Finite Element Analysis (FEA), experimentation was used to assume a vital part in finding and describing new materials. Consequently, trial research is carried out with few sample replications for an extended period, this requires a considerable need for assets and equipment. Inferable from these impediments, significant revelations about material properties and characteristics are made for the most part through human instinct [38]. In contrast, FEA as a computational tool can calculate materials' response to external forces or temperature or pressure and provides visual information on how the material will perform in real-life situations [124].

The FEA is one of the most extensively utilized methodologies in the transition from numerical analysis to ML. FEA is often combined with a different ML algorithm to simulate the responses of composite materials. For a comprehensive evaluation of reinforced composites, the pre-failure, onset of failure, and post-failure responses must be modeled. The finite element model is prepared in a Computer-Aided Design (CAD) system which involves creating computer models of the reinforced composite structures defined by geometrical parameters. Designers may examine items in several forms and evaluate them by replicating real-world settings using CAD technologies. Additional procedures may be applied in local regions before mesh formation to regulate local mesh density to make the analysis conform to real demands and enhance computational performance Okafor et al., [126]. On the upshot, FEA breaks the initial problem into manageable bits known as finite elements and then assembles the basic equations that describe each element into a more sophisticated equation that would explain the entire problem. As a consequence, FEA may interpolate a set of points using equations to construct a surface that includes all of them. X-ray images of reinforced composite microstructure can also be converted into Finite Element meshes where the geometry of all constituents and flaws are represented explicitly. In Fig. 15, Creveling et al. [125], showed how to transform tomographic image data into Finite Element meshes.

Silva et al. [51], reported an integrated approach for fatigue life prediction of adhesively bonded joints, their ML model (extremely randomized trees algorithm) was fed with stress ratio, stress amplitude, stress concentration factor, and multiaxiality which are obtained from FEA. The load-displacement data of material has been utilized successfully to train the Artificial Neural Network material model deployed in FE simulations [127]. Combined simulation and ML in the design and manufacturing of Thermoplastic Composites. The training data for ML methods were generated from parametric finite element studies. The visualization into different quality domains qualifies their approach as decision support. Ali et al. [128], implemented a Deep Convolutional Neural Network (DCNN) using data from two different types of reinforcement segments from μ-CT images. Scanned CT images, as well as pictures taken from computer-generated digital objects of the reinforcements, were used to train the neural network. In comparison to the traditional approach, the trained deep neural network was apt to provide quick and effective forecasts of various traits of the reinforcements.

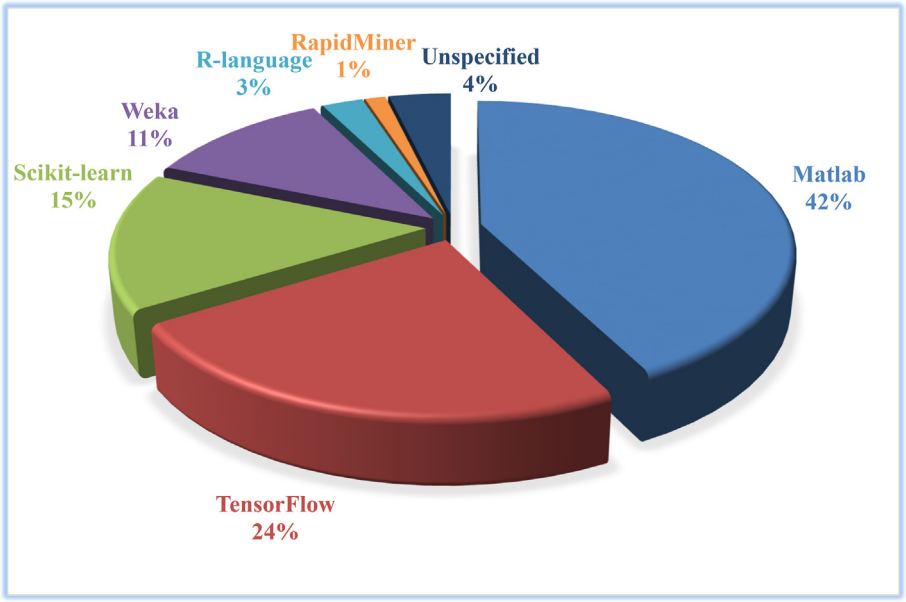


Fig. 19. Distribution of software tools used by scholars for implementation of ML **algorithms in reinforced composite design from 2018–2022.**

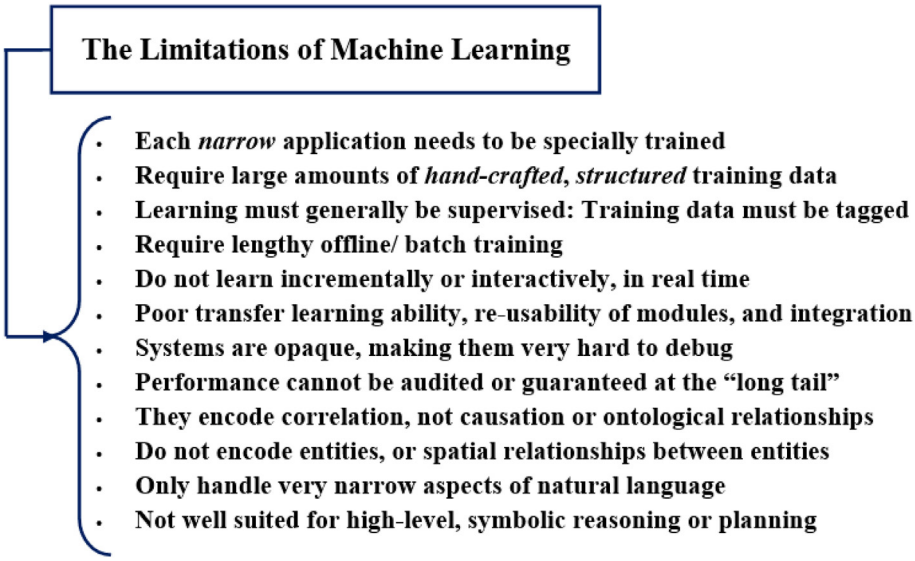


Fig. 20. The limitations of machine learning.

At the nano- and continuum scales, the finite element method can be used to simulate a wide range of mechanical behaviors of materials. In general, simulating a material's properties to predict its characteristics is much faster and less expensive than synthesizing, manufacturing, and testing it in a lab. Furthermore, simulations provide extremely fine control over surroundings as well as more thorough information on material behavior and associated mechanisms under a variety of situations, many of which are impossible or extremely difficult to detect using experiments. For example, Pathan et al. [129], used supervised ML to predict the mechanical properties of unidirectional fiber composites. Their computational framework was predicated on one thousand, eight hundred empirical volume elements that depict cylindrical fibers in a continuous phase filled in the x-direction. Their model had a high degree of accuracy in predicting the normalized properties of the random nanostructure.

Liang et al. [130], developed and trained a deep ML model that takes FEA input and outputs stress distributions directly. In the Von Mises stress distribution and peak Von Mises stress, the trained deep learning model successfully predicted the stress distributions with average errors of 0.492% and 0.891%, respectively. Finite-element discretization of partial differential equations (PDEs) that resolve quantities of interest within a machine-learning framework was introduced by Brevis et al., [131]. The authors employed an artificial neural network to create the parametric family of test spaces and found that even on very coarse meshes, they could gain a better approximation of variables of interest.

9.2. *Experimentation*

A selection of data is essential for proper training of a ML model to create an efficient ML model. The chosen subset must be accurately identifiable and representative of the complete analytical domain. It should not overlook any significant characteristics that may influence the material's final behavior. Many design parameters influence the mechanical characteristics of reinforced polymer composites, necessitating a

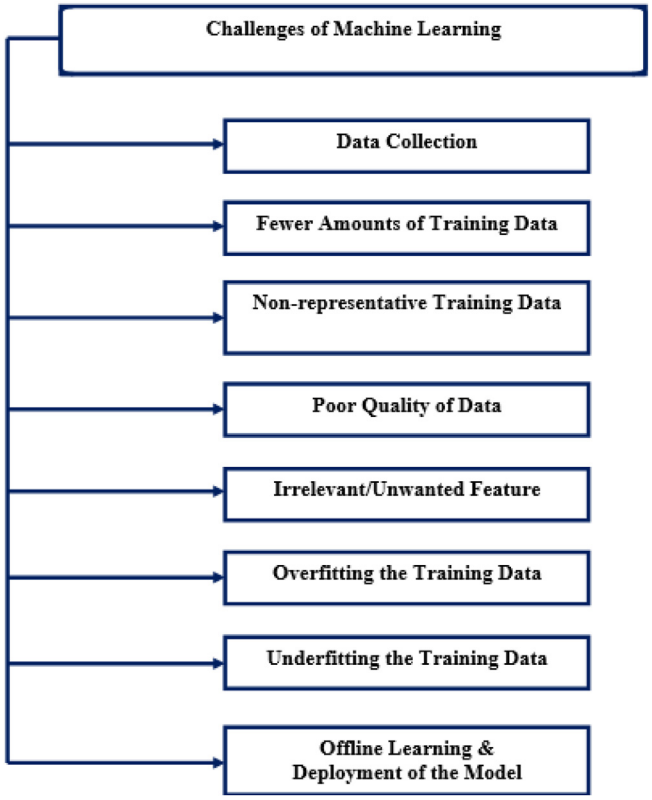


Fig. 21. Challenges of machine learning.

data-driven modeling technique to anticipate the composite's important properties. To this end, surrogate models of material characteristics and performance predictions are usually created using ML techniques and existing materials data. ML was used by Wang et al. [132], to predict the mechanical characteristics of braided-textile reinforced tubular constructions. They derived the nine input parameters for the ANN model straight from the experiments. Although the relative error of marginal sample data is high, the training and test sets' mean absolute errors are only approximately 5% and 10%, respectively. Their research shows how ML approaches may be used to anticipate overall mechanical characteristics and guide composite material design.

Alam et al. [134], studied the hardness behavior of sintered Al/SiC composites using RSM and ANN. Their findings showed that the ANN model predicted surface hardness values near the experimental findings. They established that the developed models can be used to predict the hardness of the surface within the investigation range. The effectiveness of ML algorithms was demonstrated by Qiu et al. [119], in estimating the translaminar R-curve of composites based on a basic tensile test of pre-cracked data. Their findings demonstrated that employing a machine learning-based framework to extract additional information about composites from simple experimental data is feasible.

As demonstrated in Fig. 16 by Mishra et al. [133], the load versus deflection curve for all beam typologies was consistent with the values predicted using the ANN model, implying that the ANN model can predict the experimentally recorded data with utmost precision. Liu et al. [135], observed that ANN is the ML paradigm that has gained the most traction in recent years. However, authors noted that there are still several unresolved problems that prevent the use of ANN models in the actual design and analysis of composite materials and structures, despite the fact that numerous ANN models have been employed in the design and analysis of composite materials and structures. Additionally, the data-based design paradigm is facing new opportunities and problems as a result of routine demand for large data set. This is because massive data has been shown by researchers to reduce estimation variance, which improves predictive performance.

To better understand the behavior of cotton fiber/polyvinyl chloride composites, Kazi et al. [136], developed Artificial Neural Network models based on experimental data to forecast load-displacement curves. Their method proved to be highly efficient, reducing the time and effort required to conduct behavioral research on a large number of samples, and it will aid materials designers in designing more successful studies in the future. ChemNet Deep Neural Network was created and applied by Goli et al. [137], for advanced composites design and manufacture. The machine learning-based design technique gave very accurate predictions and support for matrix phase cure kinetics, which has the potential to solve problems in thermosetting polymer design and process management.

Kharb et al. [138], deployed a Support Vector Machine (SVM) and Multi-linear Regression ML approach coupled with experimental data via Taguchi's methodology to analyze and predict the effect of reinforced silicon carbide particle size on the erosion behavior of silicon carbide reinforced polymer composites. Wang et al. [139], reported an experimental investigation combined with a random forest ML approach for the study of the time-frequency patterns of acoustic emission signals under different cutting speeds and fiber orientations. Najjar et al. [50], deployed a micromechanics model, finite element simulation of micro-indentation, and ML to predict the mechanical properties of Cu–Al_2O_3 nanocomposites. The authors obtained mechanical properties that were in very good agreement with the experimental ones achieving a 0.99 coefficient of determination R^2 for the yield strength.

9.3. *Materials databases/literature*

The number and quality of the datasets used for training are critical in creating accurate ML models. The first phase in the ML process is to collect the training dataset, which will, of course, be significantly influenced by the study's purpose. Fig. 17 shows data handling stages during the ML execution process. Many materials databases, such as AFLOW, ASM International, Materials Project, Material Connexion, MATDAT, UL Prospector, MatWeb, ChemSpider, MatMatch, Matdat, MakeItForm, and MatNavi, have been created as a result of the development of high-throughput computational materials design. These databases contain a massive amount of material attributes, including mechanical properties, derived from actual observations and first-principles computations. Free datasets may be found on sites like Kaggle, the UC-Irvine ML Repository, the Inorganic Crystal Structure Database, the Cambridge Structural Database, and the Crystallography Open Database, as well as government websites that collect and encourage open data [140]. Sub-datasets of certain qualities or material classes may be filtered and gathered successfully using the web platforms of such databases, depending on the mechanical difficulties of interest.

Examining the literature will provide labeled datasets. The amount of gathered kinds of literature in the respective field has a significant impact on the size of the collected dataset. When combined with an active ML loop, tiny datasets containing tens to hundreds of data points are appropriate for optimization techniques. Le [96] used a database of 11 input variables obtained from the literature to construct a viable Machine Learning-based model for the prediction of tensile strength of polymer carbon nanotube composites. Using data mining techniques, Ghanem and Elgazzar [97] predicted the behavior of reinforced concrete columns restricted by fibre-reinforced polymers. The experimental findings of sixty-four concrete columns under compressive concentric load are accessible in the database utilized in their analysis. ML algorithms can properly estimate the column's compressive stress and strain with low error and high accuracy, according to the authors.

Hasan et al. [52], gathered information from the literature on the effects of material characteristics, processing procedures, heat treatment, and tribological test variables on tribological properties and developed two-parameter correlations. The data was utilized to create ML models that can predict the tribological behavior of reinforced composites based

on material factors and test circumstances. By establishing links between material and tribological qualities, such a data-driven approach provided new insights into tribological processes. Iqbal et al. [141], evaluated the possible factors that might impact the durability of glass fiber reinforced polymer bars in a hostile alkaline environment, particularly saltwater and sea sand concrete, based on a complete literature data search. ML methods including artificial neuro-networking, gene expression programming, and adaptive neuro-fuzzy inference system were utilized to process the data. The trained models were put through statistical tests, and the findings revealed that they ensured a realistic estimate of tensile strength retention.

10. Property prediction in reinforced composite development using ML algorithms

In today's world, significant advancements in ML are achieved not through conventional computer scientists, but also through specialists in plenty of other industries who use these technologies to gain sustainable competitive advantage. For different purposes, various optimization approaches, including greedy algorithms and gradient-based algorithms, are commonly used to design composites and other design challenges. Nonetheless, since composite design problems are frequently nonconvex, the best solutions achieved using those optimization methods are frequently influenced by the initial geometry used in the optimization process. Hence, the solutions not only differ from one initial geometry to the next, but they can also become trapped in suboptimal local minima in some circumstances (or critical points). ML, on the other hand, is a potential alternative method for a composite design that is data-driven. In the subject of computational mechanics, recent fast improvements in ML techniques have shown a variety of useful applications.

Designers of reinforced composite product have used various ML algorithms over the last half-decade to adjust the composition and structure of composites, as well as forecast and enhance mechanical qualities that are critical to a variety of cutting-edge materials. For example, Gu et al. [142], introduced a new technique for designing various levelled materials that combined ML with a database of tens of billions of FEA models, as well as a self-learning algorithm for finding optimal materials that phased out imperfect designs in favor of superior candidates. According to the findings, which have been validated by additive manufacturing and testing, the process may produce micro structural patterns that lead to tougher and stronger materials. Xu and Gupta [143] employed an artificial neural network to broaden the transition of observed viscoelastic characteristics to anisotropic materials to construct the storage modulus master relationship in three in-plane directions. The elastic modulus is calculated after predicting the stress response with a certain strain history. The transformation can anticipate anisotropic material behavior across a wide range of temperatures and strain rates, according to the authors.

Zhang and Yokoyama [144] used an Artificial Neural Network to determine the strength ratio of laminated composite material, and a genetic algorithm was presented to improve the search process at the architecture, parameters, neural network connections, and active functions levels. The method's results are compared to those produced using lamination theory and failure criteria from the past. The results indicate that, rather than using traditional lamination and associated failure theory, an artificial neural network may be used to determine the strength ratio for in-plane loads. An artificial neural network (ANN) model for forecasting the residual strength of carbon reinforced composites (CFRCs) following a low-velocity impact was reported by Yang et al., [145]. The authors created a finite element (FE) model to predict the low-velocity impact and compression-after-impact reactions of CFRCs, which was used as input data for the back-propagation learning technique. The constructed ANN-based model proved successful in lowering experimental expenses and shortening calculation time.

Ji et al. [146], investigated the mechanical properties of carbon-reinforced PEEK-titanium hybrid laminates at various temperatures. At room temperature, 80, 150, and 220 °C, the tensile strength, flexural strength, and inter-laminar shear strength of carbon fiber-reinforced PEEK-titanium hybrid laminates with varied stacking architectures were examined. The authors created a three-layer back--propagation neural network to predict the mechanical behaviors of carbon fiber-reinforced PEEK-titanium hybrid laminates. Their findings revealed that trained back-propagation neural networks could accurately predict the mechanical performances of carbon fiber-reinforced PEEK--titanium hybrid laminates.

Using neural networks to estimate the objective function and analytical sensitivities concerning design variables as a replacement for finite element analysis, Xu et al. [147], proposed a machine learning-based approach for optimizing fiber orientations of variable stiffness carbon fiber reinforced plastic structures. The results demonstrated that as compared to the traditional single-initial design method, the suggested ML-based methodology enhances structural performance by 12.62%. Back propagation neural networks, extreme learning machines, and support vector machine algorithms were utilized by He et al. [99], to examine delamination parameters, with a specific focus on interface prediction. A theoretical model of a fiber-reinforced polymer beam with delamination under vibration was built to study how the delamination affects the frequencies and to produce a database of frequency shifts versus delamination parameters to be utilized in delamination prediction ML techniques. The support vector machine offers the highest prediction performance among the three ML algorithms, with high prediction accuracy and just a small number of samples required, according to numerical and experimental verification findings.

Gu et al. [8], used ML in a composite system and demonstrated its ability to predict mechanical parameters such as toughness and strength correctly and quickly. They employed ML approaches to develop optimum designs with attributes orders of magnitude better than the mean properties of the input training data, and at a fraction of the cost of exhaustive methods. Furthermore, even if this knowledge is lost during the training phase, the technique can recreate the detailed performances of the designs. The findings highlight machine learning's capacity to find optimal designs with relatively little training data. Dabetwar et al. [148], used signal processing, deep learning algorithms, and optimization theory to create a structural health monitoring system for smart composite structures. The hyper parameters' optimal values were determined using Bayesian optimization theory. Data augmentation was also engaged ensuring that there was sufficient data for the impact characterization model to train on. The performance of each improved neural network model was tested by comparing the test errors under each applied circumstance, according to the authors.

For detecting deterioration in composites, data-driven strategies have shown to be the most effective. Dabetwar et al. [148], used data fusion and data augmentation using deep neural networks to diagnose fatigue damage in composites. The authors found that combining artificial intelligence approaches with Lamb wave measurements can significantly enhance defect diagnosis in complicated materials like composites. Yanamandra et al. [149], suggested a reverse engineering strategy for ML modeling employing data from CT scans and SEM pictures. The authors employed imaging techniques and ML to reverse engineer a composite material component, capturing not only the geometry but also the 3D printing tool path, which was recreated using microstructure ML. The reverse engineered model has a dimensional accuracy of only 0.33% difference. Sun et al. [150], used a Fully Convolutional Neural Network (CNN) adapted from Stress Net to predict the stress field in 2D slices of segmented tomography images of a fiber-reinforced polymer specimen using a non-linear finite element (FE) simulation. Fig. 18 depicts the stress field over the complete sample volume of the composite where data produced from FE simulations of the precise microstructure was used to train and assess the network. The results of the tests reveal that the trained network successfully captures the stress distribution features, particularly on fibers, purely from segmented microstructure pictures.

Thankachan et al. [151], demonstrated how to forecast and analyze

dry sliding wear rates on innovative copper-based surface composites using ML models and statistical methodologies. A synthetic neural network-based feed forward back propagation model with topology 4-7-1 was created, and the prediction profiles matched the experimental results well. These findings suggested that using ML methodologies to perform rapid computer modeling for composite materials has potential and that the trained model may also be used to discover likely damage locations in fiber-reinforced polymers.

Using artificial intelligence, Wiangkham et al. [152], estimated the effect of loading rate and sugarcane leaf concentration on fracture toughness of sugarcane leaves and epoxy composite. The modeling employs three artificial intelligence models: artificial neural networks, extended regression neural networks, and Gaussian process regression, all of which were applied to data from a small set of 27 units. In terms of the prediction model's confirmation, it was discovered that when the amount of the predictive elements varied, the performance of the three models decreased, but remained within an acceptable range.

This section has demonstrated that ML methods are now frequently applied in a variety of polymer composite study fields. Scholars were able to forecast a variety of multifunctional features for polymeric composites depending on both the constituents and their ratios. Also, a variety of ML algorithms have been created based on the sorts of properties and datasets available. The prediction of relationships between material characteristics using high-throughput screening techniques can aid in the search for new materials. Also, predicting changes in material characteristics can direct performance design and lower the cost of research and development. Unfortunately, many of the ML algorithms have not been applied to the study of composite materials. Additionally, instead of focusing on the total mechanical characteristics of reinforced composite components, the majority of these research tend to focus on the prediction of one or two mechanical qualities. Although there are several machine learning algorithms to choose from, each one has its own limitations and range of potential uses. Also, the limitations of many variables, which lead to increasing complexity and uncertainty brought on by the unpredictability of the data, limit the scope of the majority of investigations.

11. Appraisal of emerging digital tools and platforms used for implementation of ML algorithms

Researchers are constantly investigating the growing capabilities of ML software and algorithms in the pursuit of making advancements in composites technology. There are a plethora of new digital tools and platforms that may be used to deploy various ML algorithms. ML tools include functionalities for loading data, filtering data, selecting features, classification/clustering techniques, and data visualization, among other things. MATLAB, WEKA, Orange, ELKI, MOA, KNIME, Rapid Miner, Tensor Flow, R-language, and Scikit-learn are some of these tools. Some of these tools enable deep learning algorithms while others provide a variety of ML techniques. Some are free, while others are for a fee. These software tools are written in a variety of languages and have unique features. Tables 2–8 summarised previous studies that used ML software and algorithms in reinforced composite design.

During the period covered by the present review, researchers have predominantly utilized MATLAB, TensorFlow, Scikit-learn, Weka, and R-language in that order. MATLAB is a computer application that combines an incremental design phase environment with a programming language that supports matrix and array mathematics directly. The vast number of researchers (42%) chose MATLAB to apply ML, as shown in Fig. 19. The high-performance multi-paradigm environment for numerical computation may be linked to MATLAB's widespread use in ML implementation.

TensorFlow was used in 24% of the examined articles on machine learning-aided reinforced composite design. TensorFlow is a Google deep learning toolkit that may be used for image recognition, neural networks, pattern decoding, and correlations, among other things. Tensor Flow can be simply utilized using the Anaconda tool or with Docker images. In 15% of the publications examined, Scikit-learn was utilized. Scikit-learn makes use of the matplotlib package to provide a variety of graphical data representations. Scikit-learn is a Python package that is available for free. It is designed in such a manner that it can easily operate with Python numerical and scientific libraries such as Numpy and Scipy. Both supervised and unsupervised learning is supported by Scikit-learn. Other programs can generate graphical representations as well, although Scikit-learn, Weka, and MATLAB are superior.

Weka was used in 11% of the peer-reviewed papers on ML design of reinforced composite. Weka is a suite of Java-based tools and a collection of open-source techniques for data mining and ML. By using criteria artificial intelligence systems for a combination of developmental and material challenges, the WEKA experience is changing the stability [203]. WEKA, with its user-friendly interface, maybe the easiest program to start with for a novice user. Weka has a graphical user interface (GUI) that simplifies a lot of the activities. Importing a file and displaying the variables on a scatter plot is just a few clicks away, and anybody who has used a language like R or Python to perform the same operation can relate to how many lines of code they have to write to accomplish the same task with minimal errors every time.

R-language was used in 3% of the publications that were examined. For statistical modeling and analysis, R-language is the most widely used computer language. Objects in R-language are kept in physical memory. It differs from other programming languages such as Python. Other programming languages, such as MATLAB and Python, are significantly slower than R-language. Algorithms in R-language are split up into several packages. Algorithm implementation may be tricky for programmers who are unfamiliar with packages. RapidMiner was used in 1% of the publications that were evaluated. RapidMiner is a free and open-source data and text mining software application. RapidMiner integrates data preparation, ML, deep learning, text mining, and predictive analytics into a single environment. Basic data cleansing is a challenge that ML software tools may be able to handle, although it's unclear how RapidMiner accomplishes this.

12. Drawbacks, challenges, and limitations of ML for reinforced composite technology

ML is one of the most useful artificial intelligence technologies which facilitates robust design of reinforced composite materials and enables the development of other automatic learning systems that can make judgments without continual programming. But everything in this world has both good and bad sides, in the same vein, ML has huge prospects but also has certain drawbacks, challenges, and limitations that need to be overcome.

First and foremost, the application of machine learning necessitates the availability of data that is full, consistent, reliable, and plentiful. One of the most common criticisms of ML approaches is that the black-box algorithms used to solve issues with reinforced composite materials design do not include the governing physical principles, resulting in poor interpretability. Additionally, Physic-chemical reactions, as well as chemical structure of the constituents of reinforced composite materials may not be appropriately defined to ML algorithms, which then causes to ineffective estimations. One should represent the molecular structure, atomic energy levels, and atomic forces so as to understand the composite structure arrangements in a such way that the predictions can achieve the best without implementing further experimental studies. This is because, without performing the mentioned characteristics throughout ML processes, the result may not be sufficiently taken into account. Only by understanding the causality between the model's inputs and outputs can such situations be foreseen and avoided.

The two most frequent problems encountered during machine learning-aided design of reinforced polymer composite materials are overfitting and underfitting [204]. A model is said to be overfit when it becomes so familiar with the detail and noise in the training data that it performs poorly on fresh data. In other words, the model picks up on the

noise or random oscillations in the training data and learns them as ideas. The issue is that these ideas don't apply to fresh data, which impairs the models' capacity for generalization. Nonparametric and nonlinear models, which have greater adaptability while learning a target function, are more susceptible to over fitting [205–207]. As a result, many nonparametric ML methods additionally incorporate parameters or strategies to restrict and limit the amount of detail the model learns. The exact opposite of overfitting is underfitting. ML model loses accuracy if it is trained with insufficient quantities of data, which results in the model's output being partial and erroneous. Underfitting takes place when the machine learning model is too basic to comprehend the fundamental structure of the data. This typically occurs when material designers attempt to develop a linear model using non-linear data and have a limited amount of data available in the data collection. In these situations, the ML model's complexity is destroyed, the rules are too simple to be applied to this data set, and the model also begins making incorrect predictions.

Choosing a machine-learning technique for a specific design job is tricky because of a large number of algorithms readily accessible. The limited computer knowledge possessed by most material designers has been a big problem limiting the massive deployment of machine learning in reinforced composite technology. Even though there are numerous databases for fiber composites, there are drawbacks such as the inability to easily access such data resources, inconsistencies between data generated by different groups, and inadequacies of the contemporary database, other limitations of machine learning are summarised in Fig. 20.

One of the most important components is data preparation, which is a must for any machine learning operation. For this reason, the majority of the work must be put into the data preparation phase, as machine learning will no longer be able to produce accurate results if the data is not sufficiently prepared. ML methods may not always be advantageous when dealing with material issues if the overall cost of training and design procedures is higher than normal approaches.

Another important challenge that material engineers may face when using machine learning-aided design of polymer composite is in respect of complexity of the machine learning process [208]. Additionally, the use ML entails data analysis, bias removal, data training, application of advanced mathematical calculations, etc., making the process appear more difficult and time-consuming.

A complicated and expanding problem in the use of ML such as lack of reproducibility is made worse by a lack of model testing procedures and open-source code. Reproducibility refers to the ability to run the algorithm continuously on specific inputs and get the same/very similar results [209]. Not only does reproducibility guarantee accurate designs, it also promotes transparency and offers the designer confidence in the ability to comprehend exactly what was done. Reproducibility generally lowers the possibility of mistakes, boosting the dependability of an experiment. Despite this generally held belief, a reproducibility issue is affecting numerous industries, including machine learning.

Despite several advancements in machine learning-based design of reinforced composite products, ML models still struggle to generalize situations that are different from the ones they experienced during training. ML models have difficulty learning their knowledge from one set of conditions to another. This implies that whatever a model does for a certain use case will only apply to that use case. Designers of reinforced polymer composite products are therefore compelled to continually devote resources to training new models, even though the use cases are quite identical. Other challenges of machine learning on the one hand are summarised in Fig. 21.

13. Gaps and future scope

To create predictions and analyses, most ML algorithms rely heavily on data correlation. Material designers, on the other hand, may prefer to respond to explicit and reliable causality relationships produced by logical reasoning on genuine and obvious data. Transitioning from issue solving by data correlation to problem-solving by logical reasoning might be one of the earnest needs of material engineers. There is a need for a benchmark dataset for consistent evaluation of new algorithms in ML model building and testing. As is standard practice in computer and statistical sciences, these benchmark datasets will complement the current ethos of publicly sharing ML code and data.

Understanding the ML model's area of application, interpretability, and usage for outlier identification are still big issues, which are only going to get more difficult as complicated neural network-based techniques gain popularity. While the reviewed researches show how ML may be used to create new materials with desired characteristics, they largely focused on using ML models to forecast material qualities rather than developing intelligent composite materials with desired features. Future research efforts could create new ways for using our existing understanding of reinforced composite design to inform ML and, thus, overcome some of the technology's inherent limitations.

The promise of employing ML in mechanical material design has yet to be completely realized, with several opportunities and problems to be investigated and solved. ML algorithms remain primarily niche tools that lack the robustness and generalizability seen in human learning. While previous attempts to develop prototype ML with distinct qualities such as lifetime learning and robust representation schemes have been made, the availability of human-like learning capabilities is not certain. Nevertheless, machine learning-assisted design of reinforced composite materials will continue to develop as massive and complicated as current computer resources allow, but they will always rely on specialized training, which will somehow difficult to sustain as design jobs and settings change.

14. Conclusions

This study reviewed current advancements in the use of several ML techniques in the field of reinforced composite design. The review covered the background of the reinforced composite material part design process, the role of data-driven material design, the drawbacks of traditional methods of reinforced composite development, ML protocol in composite materials technology, implementation of ML algorithms in reinforced composite material design, a general classification of ML algorithms, data hygiene and the integration of ML algorithms in material and process selection. The review of the literature as well as practical perspectives on various critical aspects of ML about reinforced composite design will open up a plethora of opportunities for researchers to investigate optimal factor relationships and discover intelligent composite materials.

Various drawbacks of traditional methods of reinforced composite development were highlighted. The success of experimental measurement depends on the quality of test facilities, the exploratory climate, and the designer's experience. Hence experimental research may be costly and time-consuming, especially when many trials are required to examine each material variable. Finite element analysis/high-throughput simulations, experimentation, and materials databases/literature were identified as the three major sources of data for the machine learning-based design of reinforced composite.

Data scientists and material engineers now have a variety of options for how they develop models to meet diverse ML needs in the field of reinforced composite design. Consequently, the review appraised popular digital tools and platforms used by various researchers for the implementation of ML algorithms in the past five years. During the period covered by the present review, researchers have predominantly utilized MATLAB, TensorFlow, Scikit-learn, Weka, and R-language in that order. The limited computer knowledge possessed by most material designers might affect their choice of ML software and the massive deployment of ML in reinforced composite technology. With the widespread use of reinforced composite materials in the industrial landscape, there is an increasing need to create innovative methods for designing and fabricating complicated composite structures. More breakthroughs

in reinforced composites design are possible shortly if researchers can deploy ML protocols in the formulation of new material parameters.

It is necessary to encourage reinforced composite researchers to make correct data from their research efforts available to advance the field of ML in fiber composite research. ML, when paired with FEA, can effectively and reliably generalize and approximate predictions of fiber characteristics as a function of composition, dramatically speeding up the development of novel composite formulations with customized attributes and functions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] Y. Fadil, S.C. Thickett, V. Agarwal, P.B. Zetterlund, Synthesis of graphene-based polymeric nanocomposites using emulsion techniques, Prog. Polym. Sci. 125 (2022), 101476.

[2] H. Gu, C. Liu, J. Zhu, J. Gu, E.K. Wujcik, L. Shao, N. Wang, H. Wei, R. Scaffaro, J. Zhang, Z. Guo, Introducing advanced composites and hybrid materials, Advanced Composites and Hybrid Materials 1 (1) (2018) 1–5.

[3] S.N.A. Safri, M.T.H. Sultan, M. Jawaid, K. Jayakrishna, Impact behaviour of hybrid composites for structural applications: a review, Compos. B Eng. 133 (2018) 112–121.

[4] C.E. Okafor, S. Iweriolor, O.I. Ani, A. Nürettin, G.O. Ekwueme, P.C. Ugwu, E.C. Nwanna, A.C. Onovo, Biobased hybrid composite design for optimum hardness and wear resistance, Composites Part C: Open Access (2023), https://doi.org/10.1016/j.jcomc.2022.100338.

[5] C.E. Okafor, D.U. Okpe, O.I. Ani, U.C. Okonkwo, Development of carbonized wood/silicon dioxide composite tailored for single-density shoe sole manufacturing, Mater. Today Commun. (2022), https://doi.org/10.1016/j.mtcomm.2022.104184.

[6] C.E. Okafor, A.C. Onovo, P.E. Imoisili, K.M. Kulkarni, C.C. Ihueze, Optimal route to robust hybridization of banana-coir fiber particulate in polymer matrix for automotive applications, Materialia 16 (2021) 101098, https://doi.org/10.1016/j.mtla.2021.101098.

[7] C. Furtado, L.F. Pereira, R.P. Tavares, M. Salgado, F. Otero, G. Catalanotti, A. Arteiro, M.A. Bessa, P.P. Camanho, A methodology to generate design allowable composite laminates using machine learning, Int. J. Solid Struct. 233 (2021), 111095.

[8] G.X. Gu, C.T. Chen, M.J. Buehler, De novo composite design based on a machine learning algorithm, Extreme Mechanics Letters 18 (2018) 19–28.

[9] T. Banerjee, S. Dey, A.P. Sekhar, S. Datta, D. Das, Design of Alumina reinforced aluminium alloy composites with improved tribo-mechanical properties: a machine learning approach, Trans. Indian Inst. Met. 73 (12) (2020) 3059–3069.

[10] A. Milad, S.H. Hussein, A.R. Khekan, M. Rashid, H. Al-Msari, T.H. Tran, Development of ensemble machine learning approaches for designing fiber-reinforced polymer composite strain prediction model, Eng. Comput. (2021) 1–13.

[11] S. Kumar, I.M. Nezhurina, Machine learning applications for the design of new materials: a review, Industry 4.0 3 (4) (2018) 186–189.

[12] C.M. Hamel, D.J. Roach, K.N. Long, F. Demoly, M.L. Dunn, H.J. Qi, Machine-learning-based design of active composite structures for 4D printing, Smart Mater. Struct. 28 (6) (2019), 065005.

[13] Z. Song, X. Chen, F. Meng, G. Cheng, C. Wang, Z. Sun, W.J. Yin, Machine Learning in Materials Design: Algorithm and Application, Chinese Physics B, 2020.

[14] J. Sun, Y. Ma, J. Li, J. Zhang, Z. Ren, X. Wang, Machine learning-aided design and prediction of cementitious composites containing graphite and slag powder, J. Build. Eng. 43 (2021), 102544.

[15] K. Guo, Z. Yang, C.H. Yu, M.J. Buehler, Artificial intelligence and machine learning in the design of mechanical materials, Mater. Horiz. 8 (4) (2021) 1153–1172.

[16] A. Bhaduri, A. Gupta, L. Graham-Brady, Stress field prediction in fiber-reinforced composite materials using a deep learning approach, Compos. B Eng. 238 (2022), 109879.

[17] C. Kim, J. Lee, J. Yoo, Machine learning-combined topology optimization for functionary graded composite structure design, Comput. Methods Appl. Mech. Eng. 387 (2021), 114158.

[18] Y.D. Boon, S.C. Joshi, S.K. Bhudolia, G. Gohel, Recent advances in the design automation for performance-optimized fiber reinforced polymer composite components, Journal of Composites Science 4 (2) (2020) 61.

[19] J. Fleischer, R. Teti, G. Lanza, P. Mativenga, H.C. Möhring, A. Caggiano, Composite materials parts manufacturing, CIRP Annals 67 (2) (2018) 603–626.

[20] M. Zoghi (Ed.), The International Handbook of FRP Composites in Civil Engineering, CRC Press, 2019.

[21] M.R.M. Asyraf, M. Rafidah, M.R. Ishak, S.M. Sapuan, N. Yidris, R.A. Ilyas, M.R. Razman, Integration of TRIZ, morphological chart, and ANP method for the development of FRP composite portable fire extinguisher, Polym. Compos. 41 (7) (2020) 2917–2932.

[22] C.E. Okafor, E.J. Okafor, J.J. Obodoeze, C.C. Ihueze, Characteristics and reliability of polyurethane wood ash composites for packaging and containerisation applications, Journal of Materials Science Research and Reviews 1 (3) (2018) 1–10, https://doi.org/10.9734/JMSRR/2018/43847.

[23] C.I. Idumah, Phosphorene polymeric nanocomposites for biomedical applications: a review, International Journal of Polymeric Materials and Polymeric Biomaterials (2022) 1–18.

[24] R. Zhang, Z. Li, Q. Sun, G. Yu, X. Wang, L. Wu, Design and characterization of the carbon fiber tube reinforced polymer composite for full ocean depth submersibles, Compos. Sci. Technol. 217 (2022), 109074.

[25] C.I. Idumah, A.C. Ezika, V.U. Okpechi, Emerging trends in polymer aerogel nanoarchitectures, surfaces, interfaces and applications, Surface. Interfac. 25 (2021), 101258.

[26] C.I. Idumah, M. Zurina, J. Ogbu, J.U. Ndem, E.C. Igba, A review on innovations in polymeric nanocomposite packaging materials and electrical sensors for food and agriculture, Compos. Interfac. (2019), https://doi.org/10.1080/09276440.2019.1600972.

[27] S.M. Sapuan, R.A. Ilyas, M.R.M. Asyraf, Design for safety in composites, in: Safety and Health in Composite Industry, Springer, Singapore, 2022, pp. 95–113.

[28] C.C. Ihueze, C.E. Okafor, S.N. Obuka, J. Abdulrahman, U.O. Onwurah, Integrity and cost evaluation of natural fibers/HDPE composite tailored for piping applications, J. Thermoplast. Compos. Mater. XX (X) (2021) 1–27, https://doi.org/10.1177/08927057211010878. Sage.

[29] C.E. Okafor, U.C. Okonkwo, I.P. Okokpujie, Trends in reinforced composite design for ionizing radiation shielding applications: a review, J. Mater. Sci. 56 (20) (2021) 11631–11655, https://doi.org/10.1007/s10853-021-06037-3. Springer.

[30] M.J. Suriani, R.A. Ilyas, M.Y.M. Zuhri, A. Khalina, M.T.H. Sultan, S.M. Sapuan, C.M. Ruzaidi, F.N. Wan, F. Zulkifli, M.M. Harussani, M.A. Azman, A critical review of natural fiber reinforced hybrid composites: processing, properties, applications, and cost, Polymers 13 (20) (2021) 3514.

[31] A.G.N. Matuda, M.P.M. Silveira, G.S.D. Andrade, A.M.D.O.D. Piva, J.P.M. Tribst, A.L.S. Borges, L. Testarelli, G. Mosca, P. Ausiello, Computer-aided design modelling and finite element analysis of premolar proximal cavities restored with resin composites, Materials 14 (9) (2021) 2366.

[32] Y. Yu, H. Liu, K. Qian, H. Yang, M. McGehee, J. Gu, D. Luo, L. Yao, Y.J. Zhang, Material characterization and precise finite element analysis of fiber reinforced thermoplastic composites for 4D printing, Comput. Aided Des. 122 (2020), 102817.

[33] G. Kumar, A. Taber, An integral representation of fields with applications to finite element analysis of spatially varying materials, Comput. Aided Des. 126 (2020), 102869.

[34] S. Waddell, E. Law, A. Inthavong, M. Parker, D.D. Arola, Optimization and characterization of printing parameters on a novel continuous carbon fiber composite material for 3Dprinting, Undergraduate Research in MSE 1 (2020), https://doi.org/10.6069/ZDQW-PW89.

[35] N. Schiavone, V. Verney, H. Askanian, High-density bio-PE and pozzolan based composites: formulation and prototype design for control of low water flow, Polymers 13 (12) (2021) 1908.

[36] S. Konstantopoulos, C. Hueber, I. Antoniadis, J. Summerscales, R. Schledjewski, Liquid composite molding reproducibility in real-world production of fiber reinforced polymeric composites: a review of challenges and solutions, Adv. Manuf. Polym. Compos. Sci. 5 (3) (2019) 85–99.

[37] V. Mohanavel, K.A. Ali, S. Prasath, T. Sathish, M. Ravichandran, Microstructural and tribological characteristics of AA6351/Si3N4 composites manufactured by stir casting, J. Mater. Res. Technol. 9 (6) (2020) 14662–14672.

[38] J. Schmidt, M.R. Marques, S. Botti, M.A. Marques, Recent advances and applications of machine learning in solid-state materials science, npj Computational Materials 5 (1) (2019) 1–36.

[39] C. Breite, A. Melnikov, A. Turon, A.B. de Morais, C. Le Bourlot, E. Maire, E. Schöberl, F. Otero, F. Mesquita, I. Sinclair, J. Costa, Detailed experimental validation and benchmarking of six models for the longitudinal tensile failure of unidirectional composites, Compos. Struct. 279 (2022), 114828.

[40] Y. Xiang, K. Shimoyama, K. Shirasu, G. Yamamoto, Machine learning-assisted high-throughput molecular dynamics simulation of high-mechanical performance carbon nanotube structure, Nanomaterials 10 (12) (2020) 2459.

[41] P. Pattnaik, A. Sharma, M. Choudhary, V. Singh, P. Agarwal, V. Kukshal, Role of machine learning in the field of Fiber reinforced polymer composites: a preliminary discussion, Mater. Today Proc. 44 (2021) 4703–4708.

[42] J. Chen, L. Wan, Y. Ismail, J. Ye, D. Yang, A micromechanics and machine learning coupled approach for failure prediction of unidirectional CFRP composites under triaxial loading: a preliminary study, Compos. Struct. 267 (2021), 113876.

[43] C.E. Okafor, C.C. Ihueze, Strength analysis and variation of elastic properties in plantain fiber/polyester composites for structural applications, in: Tri-Dung Ngo (Ed.), Composite and Nanocomposite Materials - from Knowledge to Industrial Applications, 2020, pp. 1–23, https://doi.org/10.5772/intechopen.90890. London.

[44] P. Guo, W. Meng, M. Xu, V.C. Li, Y. Bao, Predicting mechanical properties of high-performance fiber-reinforced cementitious composites by integrating micromechanics and machine learning, Materials 14 (12) (2021) 3143.

[45] C. Zeng, L. Liu, Y. Hu, W. Bian, J. Leng, Y. Liu, A viscoelastic constitutive model for shape memory polymer composites: micromechanical modeling, numerical implementation and application in 4D printing, Mech. Mater. 169 (2022), 104301.

[46] K. Kamau-Devers, S.A. Miller, Using a micromechanical viscoelastic creep model to capture multi-phase deterioration in bio-based wood polymer composites exposed to moisture, Construct. Build. Mater. 314 (2022), 125252.

[47] X. Sun, L. Yue, L. Yu, H. Shao, X. Peng, K. Zhou, F. Demoly, R. Zhao, H.J. Qi, Machine learning-evolutionary algorithm enabled design for 4D-printed active composite structures, Adv. Funct. Mater. 32 (10) (2022), 2109805.

[48] H.E. Balcıoğlu, A.Ç. Seçkin, Comparison of machine learning methods and finite element analysis on the fracture behavior of polymer composites, Arch. Appl. Mech. 91 (1) (2021) 223–239.

[49] G.A. Lyngdoh, N.K. Kelter, S. Doner, N.A. Krishnan, S. Das, Elucidating the auxetic behavior of cementitious cellular composites using finite element analysis and interpretable machine learning, Mater. Des. 213 (2022), 110341.

[50] I.M.R. Najjar, A.M. Sadoun, G.S. Alsoruji, M. Abd Elaziz, A. Wagih, Predicting the mechanical properties of Cu–Al2O3 nanocomposites using machine learning and finite element simulation of indentation experiments, Ceram. Int. 48 (6) (2022) 7748–7758.

[51] G.C. Silva, V.C. Beber, D.B. Pitz, Machine learning and finite element analysis: an integrated approach for fatigue life prediction of adhesively bonded joints, Fatig. Fract. Eng. Mater. Struct. 44 (12) (2021) 3334–3348.

[52] M.S. Hasan, A. Kordijazi, P.K. Rohatgi, M. Nosonovsky, Triboinformatics approach for friction and wear prediction of Al-Graphite composites using machine learning methods, J. Tribol. 144 (1) (2022).

[53] G. Pilania, Machine learning in materials science: from explainable predictions to autonomous design, Comput. Mater. Sci. 193 (2021), 110360.

[54] E. Stach, B. DeCost, A.G. Kusne, J. Hattrick-Simpers, K.A. Brown, K.G. Reyes, J. Schrier, S. Billinge, T. Buonassisi, I. Foster, C.P. Gomes, Autonomous experimentation systems for materials development: a community perspective, Matter 4 (9) (2021) 2702–2726.

[55] A.E. Gongora, B. Xu, W. Perry, C. Okoye, P. Riley, K.G. Reyes, E.F. Morgan, K.A. Brown, A Bayesian experimental autonomous researcher for mechanical design, Sci. Adv. 6 (15) (2020) eaaz1708.

[56] Y. Yan, L. Wang, T. Wang, X. Wang, Y. Hu, Q. Duan, Application of soft computing techniques to multiphase flow measurement: a review, Flow Meas. Instrum. 60 (2018) 30–43.

[57] J. Cai, X. Chu, K. Xu, H. Li, J. Wei, Machine learning-driven new material discovery, Nanoscale Adv. 2 (8) (2020) 3115–3130.

[58] N.T. Mahmood, M.H. Al-Muifraje, S.K. Salih, T.R. Saeed, Pattern recognition of composite motions based on EMG signal via machine learning, Engineering and Technology Journal 39 (2A) (2021) 295–305.

[59] J. Li, R.D. Salim, M.S. Aldlemy, J.M. Abdullah, Z.M. Yaseen, Fiberglass-reinforced polyester composites fatigue prediction using novel data-intelligence model, Arabian J. Sci. Eng. 44 (4) (2019) 3343–3356.

[60] C. Sacco, A.B. Radwan, A. Anderson, R. Harik, E. Gregory, Machine learning in composites manufacturing: a case study of Automated Fiber Placement inspection, Compos. Struct. 250 (2020), 112514.

[61] J.S. Huang, J.X. Liew, K.M. Liew, Data-driven machine learning approach for exploring and assessing mechanical properties of carbon nanotube-reinforced cement composites, Compos. Struct. 267 (2021), 113917.

[62] C. Wanigasekara, E. Oromiehie, A. Swain, B.G. Prusty, S.K. Nguang, Machine learning-based inverse predictive model for AFP-based thermoplastic composites, Journal of Industrial Information Integration 22 (2021), 100197.

[63] A. Kaveh, A.D. Eslamlou, S.M. Javadi, N.G. Malek, Machine learning regression approaches for predicting the ultimate buckling load of variable-stiffness composite cylinders, Acta Mech. 232 (3) (2021) 921–931.

[64] C. Mishra, P.M.V. Subbarao, Machine learning integration with combustion physics to develop a composite predictive model for reactivity controlled compression ignition engine, J. Energy Resour. Technol. 144 (4) (2021), 042302.

[65] M.A. Hickman, Machine Learning Based Prediction and Classification of Properties of Engineered Composite Materials (Doctoral Dissertation), 2020.

[66] C. Cerbu, H. Wang, M.F. Botis, Z. Huang, C. Plescan, Temperature effects on the mechanical properties of hybrid composites reinforced with vegetable and glass fibers, Mech. Mater. 149 (2020), 103538.

[67] D. Nardi, J. Sinke, Design analysis for thermoforming of thermoplastic composites: prediction and machine learning-based optimization, Composites Part C: Open Access 5 (2021), 100126.

[68] C.C. Ihueze, C.E. Okafor, U.O. Onwurah, S.N. Obuka, Q.A. Kingsley-omoyibo, Modelling creep responses of plantain fibre reinforced HDPE (PFRHDPE) for elevated temperature applications, Advanced Industrial and Engineering Polymer Research (2022), https://doi.org/10.1016/j.aiepr.2022.06.001.

[69] Y.X. Chen, L.C. Wang, P.C. Chu, A recipe parameter recommendation system for an autoclave process and an empirical study, Procedia Manuf. 51 (2020) 1046–1053.

[70] A. Casale, J. Dettman, Composite machine learning algorithm for material sourcing, J. Forensic Sci. 65 (5) (2020) 1458–1464.

[71] M. Szarski, S. Chauhan, Composite temperature profile and tooling optimization via Deep Reinforcement Learning, Compos. Appl. Sci. Manuf. 142 (2021), 106235.

[72] J. Ikemoto, T. Ushio, Continuous Deep Q-Learning with Simulator for Stabilization of Uncertain Discrete-Time Systems, 2021 *arXiv preprint arXiv:*2101.05640.

[73] Z. Zhang, J. Shi, T. Yu, A. Santomauro, A. Gordon, J. Gou, D. Wu, Predicting flexural strength of additively manufactured continuous carbon fiber-reinforced polymer composites using machine learning, J. Comput. Inf. Sci. Eng. 20 (6) (2020), 061015.

[74] X.D. Lei, X.Q. Wu, Z. Zhang, K.L. Xiao, Y.W. Wang, C.G. Huang, A machine learning model for predicting the ballistic impact resistance of unidirectional fiber-reinforced composite plate, Sci. Rep. 11 (1) (2021) 1–10.

[75] P. Swathi, K. Pothuganti, Overview of principal component analysis algorithm in machine learning, International Research Journal of Modernization in Engineering Technology and Science 2 (1) (2020) 241–245.

[76] Y. Meng, S.N. Qasem, M. Shokri, Dimension reduction of machine learning-based forecasting models employing Principal Component Analysis, Mathematics 8 (8) (2020) 1233.

[77] J.B. Rodrigues, G.C. Vasconcelos, P.R. Maciel, Screening hardware and volume factors in distributed machine learning algorithms on spark, Computing (2021) 1–23.

[78] C. Campana, R. Leger, R. Sonnier, L. Ferry, P. Ienny, Effect of post-curing temperature on mechanical properties of a flax fiber reinforced epoxy composite, Compos. Appl. Sci. Manuf. 107 (2018) 171–179.

[79] C.E. Okafor, E.J. Okafor, K.O. Ikebudu, Evaluation of machine learning methods in predicting optimum tensile strength of microwave post-cured composite tailored for weight-sensitive applications, Engineering Science and Technology, an International Journal (2021), https://doi.org/10.1016/j.jestch.2021.04.004.

[80] T.P. Mohan, K. Kanny, Processing of high weight fraction banana fiber reinforced epoxy composites using pressure induced dip casting method, J. Compos. Mater. 55 (17) (2021) 2301–2313.

[81] R. Vijay, A. Vinod, D.L. Singaravelu, M.R. Sanjay, S. Siengchin, Characterization of chemical treated and untreated natural fibers from Pennisetum Orientale grass-A potential reinforcement for lightweight polymeric applications, International Journal of Lightweight Materials and Manufacture 4 (1) (2021) 43–49.

[82] C.C. Ihueze, N.C. Okoli, U.O. Onwurah, C.E. Okafor, The effects of chemical treatments on the mechanical properties of miscanthus X giganteus fibre, J. Eng. Appl. Sci. 21 (1) (2022) 868–878.

[83] C.E. Okafor, L.C. Kebodi, C.C. Ihueze, M.R. Sanjay, S. Siengchin, U.C. Okonkwo, Development of *Dioscorea Alata* Stem Fibres as Eco-Friendly Reinforcement for Composite Materials, Journal of King Saud University - Engineering Sciences, 2022, https://doi.org/10.1016/j.jksues.2022.02.003.

[84] A. Alshoaibi, Finite element-based model for crack propagation in linear elastic materials, Engineering Solid Mechanics 8 (2) (2020) 131–142.

[85] H. Huan, L. Liu, A. Mandelis, C. Peng, X. Chen, J. Zhan, Mechanical strength evaluation of elastic materials by multi-physical nondestructive methods: a review, Appl. Sci. 10 (5) (2020) 1588.

[86] C.E. Okafor, L.C. Kebodi, J. Kandasamy, M. May, I.E. Ekengwu, Properties and performance index of natural fiber reinforced cross-ply composites made from *Dioscorea alata* stem fibers, Composites Part C: Open Access 7 (2022), 100213, https://doi.org/10.1016/j.jcomc.2021.100213.

[87] C.A. Diniz, S.S. Cunha, G.F. Gomes, A.C. Ancelotti, Optimization of the layers of composite materials from neural networks with Tsai–Wu failure criterion, J. Fail. Anal. Prev. 19 (3) (2019) 709–715.

[88] D. Xu, P.F. Liu, Z.P. Chen, A deep learning method for damage prognostics of fiber-reinforced composite laminates using acoustic emission, Eng. Fract. Mech. 259 (2022), 108139.

[89] A. Khan, N. Kim, J.K. Shin, H.S. Kim, B.D. Youn, Damage assessment of smart composite structures via machine learning: a review, JMST Advances 1 (1) (2019) 107–124.

[90] J. Stuckner, S. Graeber, B. Weborg, T.M. Ricks, S.M. Arnold, Tractable multiscale modeling with an embedded microscale surrogate, in: AIAA Scitech 2021 Forum, 2021, p. 1963.

[91] J. Fernández, M. Chiachío, J. Chiachío, R. Muñoz, F. Herrera, Uncertainty quantification in neural networks by approximate bayesian computation: application to fatigue in composite materials, Eng. Appl. Artif. Intell. 107 (2022), 104511.

[92] J. Liu, Y. Zhang, Y. Zhang, S. Kitipornchai, J. Yang, Machine Learning Assisted Prediction of Mechanical Properties of Graphene/Aluminium Nanocomposite Based on Molecular Dynamics Simulation, Materials & Design, 2021, 110334.

[93] E. Fereiduni, A. Ghasemi, M. Elbestawi, Microstructural characterization and mechanical properties of nano-scale/sub-micron TiB-reinforced titanium matrix composites fabricated by laser powder bed fusion, J. Alloys Compd. 896 (2022), 163054.

[94] E. Ford, K. Maneparambil, S. Rajan, N. Neithalath, Machine learning-based accelerated property prediction of two-phase materials using microstructural descriptors and finite element analysis, Comput. Mater. Sci. 191 (2021), 110328.

[95] J.F. Durodola, Machine learning for design, phase transformation, and mechanical properties of alloys, Prog. Mater. Sci. 123 (2022), 100797.

[96] T.T. Le, Prediction of tensile strength of polymer carbon nanotube composites using the practical machine learning method, J. Compos. Mater. 55 (6) (2021) 787–811.

[97] S.Y. Ghanem, H. Elgazzar, Predicting the behavior of reinforced concrete columns confined by fiber-reinforced polymers using data mining techniques, SN Appl. Sci. 3 (2) (2021) 1–13.

[98] A.E. Maxwell, T.A. Warner, F. Fang, Implementation of machine-learning classification in remote sensing: an applied review, Int. J. Rem. Sens. 39 (9) (2018) 2784–2817.

[99] M. He, Y. Wang, K. Ram Ramakrishnan, Z. Zhang, A comparison of machine learning algorithms for assessment of delamination in fiber-reinforced polymer composite beams, Struct. Health Monit. 1475921720967157 (2020).

[100] C.E. Okafor, in: F.A. Anyogu, C.A. Eme, J. An Ogbodo (Eds.), Review of Natural Fiber Composite Design for Sustainable Infrastructural Development, *University-Led Knowledge And Innovation For Sustainable Development.* Nnamdi Azikiwe University Book Series on Sustainable Development, Abuja, *Boldscholar Research Ltd*, 2021, pp. 82–98.

[101] L. Kerni, S. Singh, A. Patnaik, N. Kumar, A review on natural fiber reinforced composites, Mater. Today Proc. 28 (2020) 1616–1621.

[102] D.K. Rajak, D.D. Pagar, R. Kumar, C.I. Pruncu, Recent progress of reinforcement materials: a comprehensive overview of composite materials, J. Mater. Res. Technol. 8 (6) (2019) 6354–6374.

[103] C.E. Okafor, S.C. Okoroafor, C.O. Umeasiegbu, M.C. Nwosu, K.O. Akpojaro, K.N. Nwuzor, H.C. Olechukwu, Optimization of comminution process parameters in particulate fiber development, IRE J. 2 (2) (2018) 24–34.

[104] C.E. Okafor, A.C. Onovo, O.I. Ani, C.M. Obele, D. Dziki, C.C. Ihueze, U.C. Okonkwo, Mathematical study of bio-fibre comminution process as first step towards valorization of post-harvest waste materials, Cleaner Materials 4 (2022), 100067, https://doi.org/10.1016/j.clema.2022.100067.

[105] C.E. Okafor, A.C. Onovo, C.C. Ihueze, Predictive energy requirement models in bio-fiber comminution process, SN Appl. Sci. 2 (2020) 2047, https://doi.org/10.1007/s42452-020-03781-0.

[106] X. Zhu, Z. Wan, D.C. Tsang, M. He, D. Hou, Z. Su, J. Shang, Machine learning for the selection of carbon-based materials for tetracycline and sulfamethoxazole adsorption, Chem. Eng. J. 406 (2021), 126782.

[107] D. Merayo, A. Rodriguez-Prieto, A.M. Camacho, Comparative analysis of artificial intelligence techniques for material selection applied to manufacturing in Industry 4.0, Procedia Manuf. 41 (2019) 42–49.

[108] J.S. Huang, J.X. Liew, A.S. Ademiloye, K.M. Liew, Artificial intelligence in materials modeling and design, Arch. Comput. Methods Eng. (2020) 1–15.

[109] T. Wang, M. Shao, R. Guo, F. Tao, G. Zhang, H. Snoussi, X. Tang, Surrogate model via artificial intelligence method for accelerating screening materials and performance prediction, Adv. Funct. Mater. 31 (8) (2021), 2006245.

[110] V. Teraiya, D. Jariwala, H.V. Patel, D. Babariya, Material selection of connecting rod using primary multi attribute decision making methods: a comparative study, Mater. Today Proc. 5 (9) (2018) 17223–17230.

[111] E.K. Arthur, E. Gikunoo, F.O. Agyemang, S.T. Azeko, A. Andrews, A. Twenewaa, Material selection for water pipes by the multi-objective decision-making method: the case of alternative materials for PVC pipes, J. Sci. Technol. 5 (1) (2020) 29–42.

[112] A. Afradi, I. Alavi, M. Moslemi, Selecting the most suitable method for extracting construction materials in Iran through the fuzzy multi-attribute decision-making model, J. Inst. Eng.: Series D 102 (1) (2021) 113–123.

[113] E.A. Jannatabadi, M. Goharimanesh, A. Jahan, A. Akbari, A decision-making methodology for material selection using genetic algorithm, Int. J. Inf. Decis. Sci. 11 (4) (2019) 269–299.

[114] V. Kukhar, E. Balalayeva, S. Hurkovska, Y. Sahirov, O. Markov, A. Prysiazhnyi, O. Anishchenko, The Selection of options for closed-die forging of complex parts using computer simulation by the criteria of material savings and minimum forging force, in: Intelligent Communication, Control and Devices, Springer, Singapore, 2020, pp. 325–331.

[115] G.O. Odu, Material selection optimization using weighted sum method and team-compromise instrument, Ind. Eng. Lett. 8 (4) (2018) 1–11.

[116] S. Misra, Y. Wu, Machine learning assisted segmentation of scanning electron microscopy images of organic-rich shales with feature extraction and feature ranking, Machine Learning for Subsurface Characterization 289 (2019).

[117] H. Zhang, Y. Wu, K. Wang, Y. Peng, D. Wang, S. Yao, J. Wang, Materials selection of 3D-printed continuous carbon fiber reinforced composites considering multiple criteria, Mater. Des. 196 (2020), 109140.

[118] S.H. Mousavi-Nasab, A. Sotoudeh-Anvari, A new multi-criteria decision-making approach for sustainable material selection problem: a critical study on rank reversal problem, J. Clean. Prod. 182 (2018) 466–484.

[119] C. Qiu, Y. Han, L. Shanmugam, Z. Guan, Z. Zhang, S. Du, J. Yang, Machine learning-based prediction of the translaminar R-curve of composites from a simple tensile test of pre-cracked samples, Journal of Micromechanics and Molecular Physics 6 (2021), 2050017, 01.

[120] W. Hasselbring, L. Carr, S. Hettrick, H. Packer, T. Tiropanis, From FAIR research data toward FAIR and open research software, IT Inf. Technol. 62 (1) (2020) 39–47.

[121] S. Yu, D. Zhang, K. Qian, Numerical analysis of macro-scale mechanical behaviors of 3D orthogonal woven composites using a voxel-based finite element model, Appl. Compos. Mater. 26 (1) (2019) 65–83.

[122] C.E. Okafor, C.C. Ihueze, The graft of ANN-FEM technique in macro-mechanics of multi-oriented natural fiber/polyester laminates, Mechanics of Advanced Composite Structures (2021), https://doi.org/10.22075/macs.2021.20579.1263. Semnan University.

[123] Z. Qi, N. Zhang, Y. Liu, W. Chen, Prediction of mechanical properties of carbon fiber based on cross-scale FEM and machine learning, Compos. Struct. 212 (2019) 199–206.

[124] C.E. Okafor, C.S. Metu, Theoretical fatigue response of plantain fiber based composites in structural applications, in: A. Zingoni (Ed.), Advances in Engineering Materials, Structures and Systems: Innovations, Mechanics and Applications, 2019, pp. 638–643, https://doi.org/10.1201/9780429426506-112. London.

[125] P.J. Creveling, J. Fisher, N. LeBaron, M.W. Czabaj, 4D Imaging of ceramic matrix composites during polymer infiltration and pyrolysis, Acta Mater. 201 (2020) 547–560.

[126] C.E. Okafor, O.O. Oghenemaero, M. Chukwuebuka, O.O. Isaac, Adaptive design of a universal automotive ball joint separating device, Transport Eng. 2 (2020), 100010, https://doi.org/10.1016/j.treng.2020.100010.

[127] A. Hürkamp, S. Gellrich, T. Ossowski, J. Beuscher, S. Thiede, C. Herrmann, K. Dröder, Combining simulation and machine learning as digital twin for the manufacturing of overmolded thermoplastic composites, Journal of Manufacturing and Materials Processing 4 (3) (2020) 92.

[128] M.A. Ali, Q. Guan, R. Umer, W.J. Cantwell, T. Zhang, Deep learning-based semantic segmentation of μCT images for creating digital material twins of fibrous reinforcements, Compos. Appl. Sci. Manuf. 139 (2020), 106131.

[129] M.V. Pathan, S.A. Ponnusami, J. Pathan, R. Pitisongsawat, B. Erice, N. Petrinic, V.L. Tagarielli, Predictions of the mechanical properties of unidirectional fiber composites by supervised machine learning, Sci. Rep. 9 (1) (2019) 1–10.

[130] L. Liang, M. Liu, C. Martin, W. Sun, A deep learning approach to estimate stress distribution: a fast and accurate surrogate of finite-element analysis, J. R. Soc. Interface 15 (138) (2018), 20170844.

[131] I. Brevis, I. Muga, K.G. van der Zee, Data-driven Finite Elements Methods: Machine Learning Acceleration of Goal-Oriented Computations, 2020 *arXiv preprint* arXiv: 2003.04485.

[132] H. Wang, Z. Zhang, L. Liu, Prediction and fitting of weld morphology of Al alloy-CFRP welding-rivet hybrid bonding joint based on GA-BP neural network, J. Manuf. Process. 63 (2021) 109–120.

[133] M. Mishra, A. Agarwal, D. Maity, Neural-network-based approach to predict the deflection of plain, steel-reinforced, and bamboo-reinforced concrete beams from experimental data, SN Appl. Sci. 1 (6) (2019) 1–11.

[134] M.A. Alam, H.Y. Hamdan, M. Azeem, P.B. Hussain, M.S. bin Salit, R. Khan, S. Arif, A.H. Ansari, Modeling and optimization of hardness behavior of sintered Al/SiC composites using RSM and ANN: a comparative study, J. Mater. Res. Technol. 9 (6) (2020) 14036–14050.

[135] X. Liu, S. Tian, F. Tao, H. Du, W. Yu, Machine learning-assisted modeling of composite materials and structures: a review, in: AIAA Scitech 2021 Forum, 2021, p. 2023.

[136] M.K. Kazi, F. Eljack, E. Mahdi, Predictive ANN models for varying filler content for cotton fiber/PVC composites based on experimental load-displacement curves, Compos. Struct. 254 (2020), 112885.

[137] E. Goli, S. Vyas, S. Koric, N. Sobh, P.H. Geubelle, ChemNet: a deep neural network for advanced composites manufacturing, J. Phys. Chem. B 124 (42) (2020) 9428–9437.

[138] S.S. Kharb, P. Antil, S. Singh, S.K. Antil, P. Sihag, A. Kumar, Machine learning-based erosion behavior of silicon carbide reinforced polymer composites, Silicon 13 (2021) 1113–1119.

[139] Z. Wang, F. Chegdani, N. Yalamarti, B. Takabi, B. Tai, M. El Mansori, S. Bukkapatnam, Acoustic emission characterization of natural fiber reinforced plastic composite machining using a random forest machine learning model, J. Manuf. Sci. Eng. 142 (3) (2020), 031003.

[140] S. Chibani, F.X. Coudert, Machine learning approaches for the prediction of materials properties, Apl. Mater. 8 (8) (2020), 080701.

[141] M. Iqbal, D. Zhang, F.E. Jalal, Durability evaluation of GFRP rebars in the harsh alkaline environment using optimized tree-based random forest model, J. Ocean Sci. Eng. 7 (6) (2022) 596–606.

[142] G.X. Gu, C.T. Chen, D.J. Richmond, M.J. Buehler, Bioinspired hierarchical composite design using machine learning: simulation, additive manufacturing, and experiment, Mater. Horiz. 5 (5) (2018) 939–945.

[143] X. Xu, N. Gupta, An artificial neural network approach to determine the elastic modulus of carbon fiber-reinforced laminates, JOM 71 (11) (2019) 4015–4023.

[144] H. Zhang, A. Yokoyama, Predicting strength ratio of laminated composite material with evolutionary artificial neural network, Int. J. Adv. Comput. Sci. Appl. 12 (6) (2021) 11–18.

[145] B. Yang, K. Fu, J. Lee, Y. Li, Artificial neural network (ANN)-Based residual strength prediction of carbon fibre reinforced composites (CFRCs) after impact, Appl. Compos. Mater. 28 (3) (2021) 809–833.

[146] C. Ji, J. Hu, B. Wang, Y. Zou, Y. Yang, Y. Sun, Mechanical behavior prediction of CF/PEEK-titanium hybrid laminates considering temperature effect by an artificial neural network, Compos. Struct. 262 (2021), 113367.

[147] Y. Xu, Y. Gao, C. Wu, J. Fang, G. Sun, G.P. Steven, Q. Li, Machine learning-based topology optimization of fiber orientation for variable stiffness composite structures, Int. J. Numer. Methods Eng. 122 (22) (2021) 6736–6755.

[148] S. Dabetwar, S. Ekwaro-Osire, J.P. Dias, Fatigue damage diagnostics of composites using data fusion and data augmentation with deep neural networks, Journal of Nondestructive Evaluation, Diagnostics, and Prognostics of Engineering Systems 5 (2) (2022).

[149] K. Yanamandra, G.L. Chen, X. Xu, G. Mac, N. Gupta, Reverse engineering of additive manufactured composite parts by toolpath reconstruction using imaging and machine learning, Compos. Sci. Technol. 198 (2020), 108318.

[150] Y. Sun, I. Hanhan, M.D. Sangid, G. Lin, Predicting Mechanical Properties from Microstructure Images in Fiber-Reinforced Polymers Using Convolutional Neural Networks, 2020 *arXiv preprint arXiv:2010.03675*.

[151] T. Thankachan, K.S. Prakash, V. Kavimani, S.R. Silambarasan, Machine learning and statistical approach to predicting and analyze Wear rates in copper surface composites, Met. Mater. Int. 27 (2) (2021) 220–234.

[152] A. Wiangkham, A. Ariyarit, P. Aengchuan, Prediction of the influence of loading rate and sugarcane leaves concentration on fracture toughness of sugarcane leaves and epoxy composite using artificial intelligence, Theor. Appl. Fract. Mech. 117 (2022), 103188.

[153] G.A. Lyngdoh, S. Das, Integrating multiscale numerical simulations with machine learning to predict the strain sensing efficiency of nano-engineered smart cementitious composites, Mater. Des. 209 (2021), 109995.

[154] S. Hankins, L. Kotthoff, R.S. Fertig III, Bio-like composite microstructure designs for enhanced damage tolerance via machine learning, in: American Society of Composites 34th Technical Conference, 2019.

[155] S.K. Antil, P. Antil, S. Singh, A. Kumar, C.I. Pruncu, Artificial neural network and response surface methodology based analysis on solid particle erosion behavior of polymer matrix composites, Materials 13 (2020), https://doi.org/10.3390/ma13061381.

[156] A.A. Baseer, D.V. Ravi-Shankar, M.M. Hussain, Interfacial and tensile properties of hybrid FRP composites using DNN structure with optimization model, Surf. Rev. Lett. 27 (2020), 1950099, 02.
[157] P. Antil, S. Singh, A. Manna, Analysis on the effect of electroless coated SiC p on mechanical properties of polymer matrix composites, Part. Sci. Technol. 37 (7) (2019) 791–798.
[158] A. Challapalli, G. Li, Machine learning assisted the design of a new lattice core for sandwich structures with superior load-carrying capacity, Sci. Rep. 11 (1) (2021) 1–10.
[159] H.D. Roh, D. Lee, I.Y. Lee, Y.B. Park, Machine learning aided the design of smart, self-sensing fiber-reinforced plastics, Composites Part C: Open Access 6 (2021), 100186.
[160] B. Liu, N. Vu-Bac, T. Rabczuk, A stochastic multiscale method for the prediction of the thermal conductivity of Polymer nanocomposites through hybrid machine learning algorithms, Compos. Struct. 273 (2021), 114269.
[161] S. Sherman, J. Simmons, C. Przybyla, Mesoscale characterization of continuous fiber reinforced composites through machine learning: fiber chirality, Acta Mater. 181 (2019) 447–459.
[162] B.B. Mishra, A. Kumar, P. Samui, T. Roshni, Buckling of the Laminated Composite Skew Plate Using FEM and Machine Learning Methods, Engineering Computations, 2020.
[163] M.R. Prajna, P.J. Antony, N.A. Jnanesh, Machine learning approach for flexural characterization of smart material, 1, in: Journal of Physics: Conference Series vol. 1142, IOP Publishing, 2018, November, 012007.
[164] S. Ferdousi, Q. Chen, M. Soltani, J. Zhu, P. Cao, W. Choi, R. Advincula, Y. Jiang, Characterize traction–separation relation and interfacial imperfections by data-driven machine learning models, Sci. Rep. 11 (1) (2021) 1–9.
[165] A. Singh, Z. Gu, X. Hou, Y. Liu, D.J. Hughes, Design Optimization of Braided Composite Beams for Lightweight Rail Structures Using Machine Learning Methods, Composite Structures, 2021, 115107.
[166] D. Field, Y. Ammouche, J.M. Peña, A. Jérusalem, Machine learning based multiscale calibration of mesoscopic constitutive models for composite materials: application to brain white matter, Comput. Mech. 67 (6) (2021) 1629–1643.
[167] F.H. Bhuiyan, L. Kotthoff, R.S. Fertig III, A machine learning technique to predict the biaxial failure envelope of the unidirectional composite lamina, 33rd Technical Conference of the American Society for Composites 3 (2018) 1451–1463.
[168] F. Lambiase, V. Grossi, A. Paoletti, Machine learning is applied tothe process design of hybrid metal-polymer joints, J. Manuf. Process. 58 (2020) 92–100.
[169] A. Sharma, P. Madhushri, V. Kushvaha, A. Kumar, Prediction of the fracture toughness of silicafilled epoxy composites using K-nearest neighbor (KNN) method, in: 2020 International Conference *on Computational Performance Evaluation (ComPE)*, IEEE, 2020, July, pp. 194–198.
[170] A. Mairpady, A.H.I. Mourad, M.S. Mozumder, Statistical and machine learning-driven optimization of mechanical properties in designing durable HDPE nanobiocomposites, Polymers 13 (18) (2021) 3100.
[171] H. Kumar, R.P. Swamy, Fatigue life prediction of glass fiber reinforced epoxy composites using artificial neural networks, Compos. Commun. (2021), 100812.
[172] N. Altinkök, Modeling of the mechanical and physical properties of hybrid composites produced by gas pressure infiltration, J. Braz. Soc. Mech. Sci. Eng. 41 (1) (2019) 1–16.
[173] J.O. Ighalo, C.A. Igwegbe, A.G. Adeniyi, S.A. Abdulkareem, Artificial neural network modeling of the water absorption behavior of plantain peel and bamboo fibers reinforced polystyrene composites, J. Macromol. Sci., Part B 60 (7) (2021) 472–484.
[174] R.M. Di-Benedetto, E.C. Botelho, A. Janotti, A.A. Junior, G.F. Gomes, Development of an artificial neural network for predicting energy absorption capability of thermoplastic commingled composites, Compos. Struct. 257 (2021), 113131.
[175] N. Zobeiry, J. Reiner, R. Vaziri, Theory-guided machine learning for damage characterization of composites, Compos. Struct. 246 (2020), 112407.
[176] B.B. Yin, K.M. Liew, Machine learning and materials informatics approaches for evaluating the interfacial properties of fiber-reinforced composites, Compos. Struct. 273 (2021), 114328.
[177] J. Reiner, R. Vaziri, N. Zobeiry, Machine learning assisted characterization and simulation of compressive damage in composite laminates, Compos. Struct. 273 (2021), 114290.
[178] M.K. Kazi, F. Eljack, E. Mahdi, Data-driven modeling to predict the load vs. displacement curves of targeted composite materials for industry 4.0 and smart manufacturing, Compos. Struct. 258 (2021), 113207.
[179] C.T. Chen, G.X. Gu, Effect of constituent materials on composite performance: exploring design strategies via machine learning, Advanced Theory and Simulations 2 (6) (2019), 1900056.
[180] M.K. Kazi, F. Eljack, E. Mahdi, Design of composite rectangular tubes for optimum crashworthiness performance via experimental and ANN techniques, Compos. Struct. 279 (2022), 114858.
[181] G.L. Chen, K. Yanamandra, N. Gupta, Artificial neural networks framework for detection of defects in 3D-printed fiber reinforcement composites, JOM (2021) 1–10.
[18][2]] F. Tao, X. Liu, H. Du, W. Yu, Learning composite constitutive laws via coupling Abaqus and deep neural network, Compos. Struct. 272 (2021), 114137.
[183] X. Liu, F. Gasco, J. Goodsell, W. Yu, Initial failure strength prediction of woven composites using a new yarn failure criterion constructed by deep learning, Compos. Struct. 230 (2019), 111505.
[184] M.A. Matos, S.T. Pinho, V.L. Tagarielli, Application of machine learning to predict the multiaxial strain-sensing response of CNT-polymer composites, Carbon 146 (2019) 265–275.
[185] P. Bleiziffer, J. Hofmann, R. Zboray, T. Wiege, R. Herger, Predicting the fiber orientation in glass fiber reinforced polymers using the moment of inertia and convolutional neural networks, Eng. Appl. Artif. Intell. 104 (2021), 104351.
[186] C. Yang, Y. Kim, S. Ryu, G.X. Gu, Prediction of composite microstructure stress-strain curves using convolutional neural networks, Mater. Des. 189 (2020), 108509.
[187] N. Shirolkar, P. Patwardhan, A. Rahman, A. Spear, S. Kumar, Investigating the efficacy of machine learning tools in modeling the continuous stabilization and carbonization process and predicting carbon fiber properties, Carbon 174 (2021) 605–616.
[188] V. Daghigh, T.E. Lacy Jr., H. Daghigh, G. Gu, K.T. Baghaei, M.F. Horstemeyer, C.U. Pittman Jr., Machine learning predictions on fracture toughness of multiscale bio-nano-composites, J. Reinforc. Plast. Compos. 39 (15–16) (2020) 587–598.
[189] Y. Lin, Z. Guan, The use of machine learning for the prediction of the uniformity of the degree of cure of a composite in an autoclave, Aerospace 8 (5) (2021) 130.
[190] S.K. Bhattacharya, R. Sahara, D. Bozic, J. Ruzic, A Data Analytics Approach to Predict the Hardness of the Copper Matrix Composites, 2020 *arXiv preprint arXiv: 2002.10649*.
[191] H.Q. Le, T.T. Truong, D. Dinh-Cong, T. Nguyen-Thoi, A deep feed-forward neural network for damage detection in functionally graded carbon nanotube-reinforced composite plates using modal kinetic energy, Front. Struct. Civ. Eng. (2021) 1–27.
[192] G.A. Sengodan, Prediction of two-phase composite microstructure properties through deep learning of reduced dimensional structure-response data, Compos. B Eng. 225 (2021), 109282.
[193] P.J. Creveling, J. Fisher, C. Dahlkamp, M.W. Czabaj, Modeling as-manufactured fiber-reinforced microstructures based on X-ray microcomputed tomography, Compos. Sci. Technol. 214 (2021), 109004.
[194] I. Lizarralde, E. Sapountzi, T. Bénéthuilière, F. Sket, C. González, An X-ray computed tomography analysis of damage induced by thermal cycling in non-crimp fabric composites, Compos. Appl. Sci. Manuf. 152 (2022), 106699.
[195] J. Castro, F. Sket, L. Helfen, C. González, In situ local imaging and analysis of impregnation during liquid molding of composite materials using synchrotron radiation computed laminography, Compos. Sci. Technol. 215 (2021), 108999.
[196] A. Khan, H.S. Kim, Assessment of delaminated smart composite laminates via system identification and supervised learning, Compos. Struct. 206 (2018) 354–362.
[197] I. Hanhan, R. Agyei, X. Xiao, M.D. Sangid, Comparing non-destructive 3D X-ray computed tomography with destructive optical microscopy for microstructural characterization of fiber-reinforced composites, Compos. Sci. Technol. 184 (2019), 107843.
[198] I. Hanhan, R.F. Agyei, X. Xiao, M.D. Sangid, Predicting Microstructural Void nucleation in Discontinuous fiber composites through coupled in-situ X-ray tomography experiments and Simulations, Sci. Rep. 10 (1) (2020) 1–8.
[199] A. Nair, C.S. Cai, X. Kong, Acoustic emission pattern recognition in CFRP retrofitted RC beams for failure mode identification, Compos. B Eng. 161 (2019) 691–701.
[200] E. Koumoulos, G. Konstantopoulos, C. Charitidis, Applying machine learning to nanoindentation data of (nano-) enhanced composites, Fibers 8 (1) (2020) 3.
[201] G. Konstantopoulos, E.P. Koumoulos, C.A. Charitidis, Classification of the mechanism of reinforcement in the fiber-matrix interface: application of Machine Learning on nanoindentation data, Mater. Des. 192 (2020), 108705.
[202] P.R. Pati, Prediction and wear performance of red brick dust filled glass–epoxy composites using neural networks, Int. J. Plast. Technol. 23 (2) (2019) 253–260.
[203] S.K. Srivastava, Y.K. Sharma, S. Kumar, Using of WEKA tool in machine learning: a review, International Journal of Advanced Science and Technology 29 (No. 8) (2020) 4456–4466, 2020.
[204] H. Zhang, L. Zhang, Y. Jiang, Overfitting and underfitting analysis for deep learning based end-to-end communication systems, in: 2019 11th International Conference *on Wireless Communications And Signal Processing (WCSP)*, IEEE, 2019, October, pp. 1–6.
[205] R. Sparapani, C. Spanbauer, R. McCulloch, Nonparametric machine learning and efficient computation with bayesian additive regression trees: the BART R package, J. Stat. Software 97 (2021) 1–66.
[206] B. Liu, L. Gao, B. Li, R. Marcos-Martinez, B.A. Bryan, Nonparametric machine learning for mapping forest cover and exploring influential factors, Landsc. Ecol. 35 (7) (2020) 1683–1699.
[207] X. Xu, M. Elgamal, M. Doddamani, N. Gupta, Tailoring composite materials for nonlinear viscoelastic properties using artificial neural networks, J. Compos. Mater. 55 (11) (2021) 1547–1560.
[208] F.M. Siemers, C. Feldmann, J. Bajorath, Minimal data requirements for accurate compound activity prediction using machine learning methods of different complexity, Cell Reports Physical Science (2022), 101113.
[209] A.J. Brunner, Fracture mechanics testing of fiber-reinforced polymer composites: the effects of the "human factor" on repeatability and reproducibility of test data, Eng. Fract. Mech. 264 (2022), 108340.